

SESSION 2: MEMORY WALL AND INTERCONNECT WALL

Optical Interconnect for Scalable AI Systems

Seongwon (Gabriel) Yoon, PI: Prof. Shimeng Yu

gabrielyoon@gatech.edu

Agenda

OUTLINE OF THIS TALK

GenAI LLM Era

Distributed
Infrastructure for ML

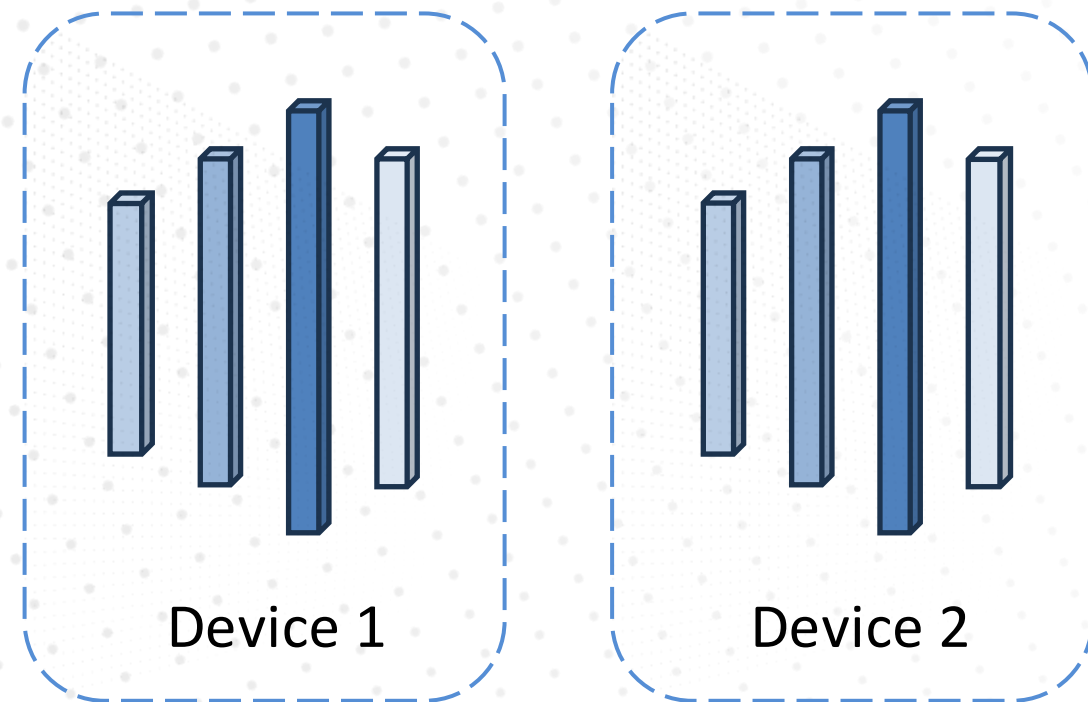
Optical Interconnect
Solutions

GenAI LLM Era

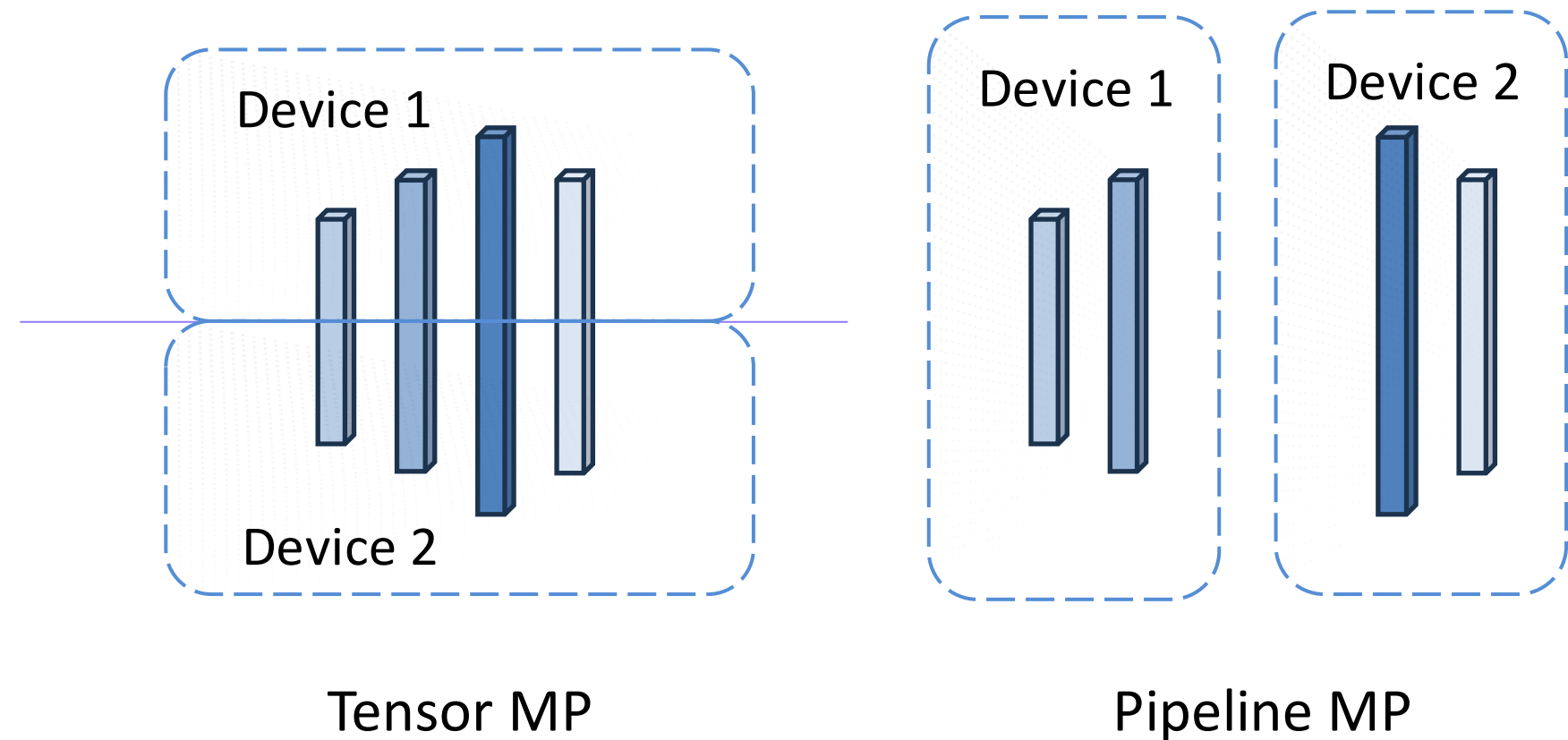
- Distributed Training
- LLM / Mixture-of-Experts
- All-to-All traffic pattern in MoE

Distributed Training: Tensor & Model Parallelism

Data Parallelism (DP)

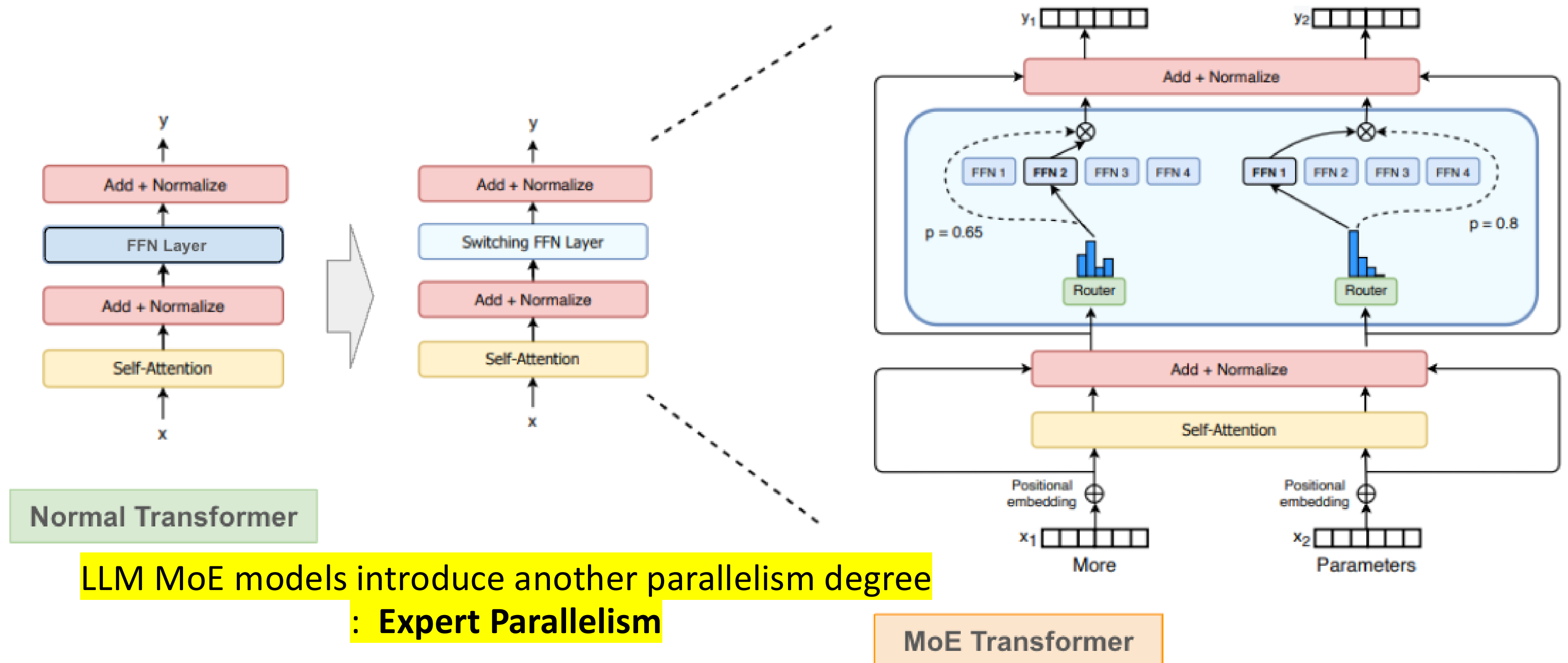


Model Parallelism (MP)

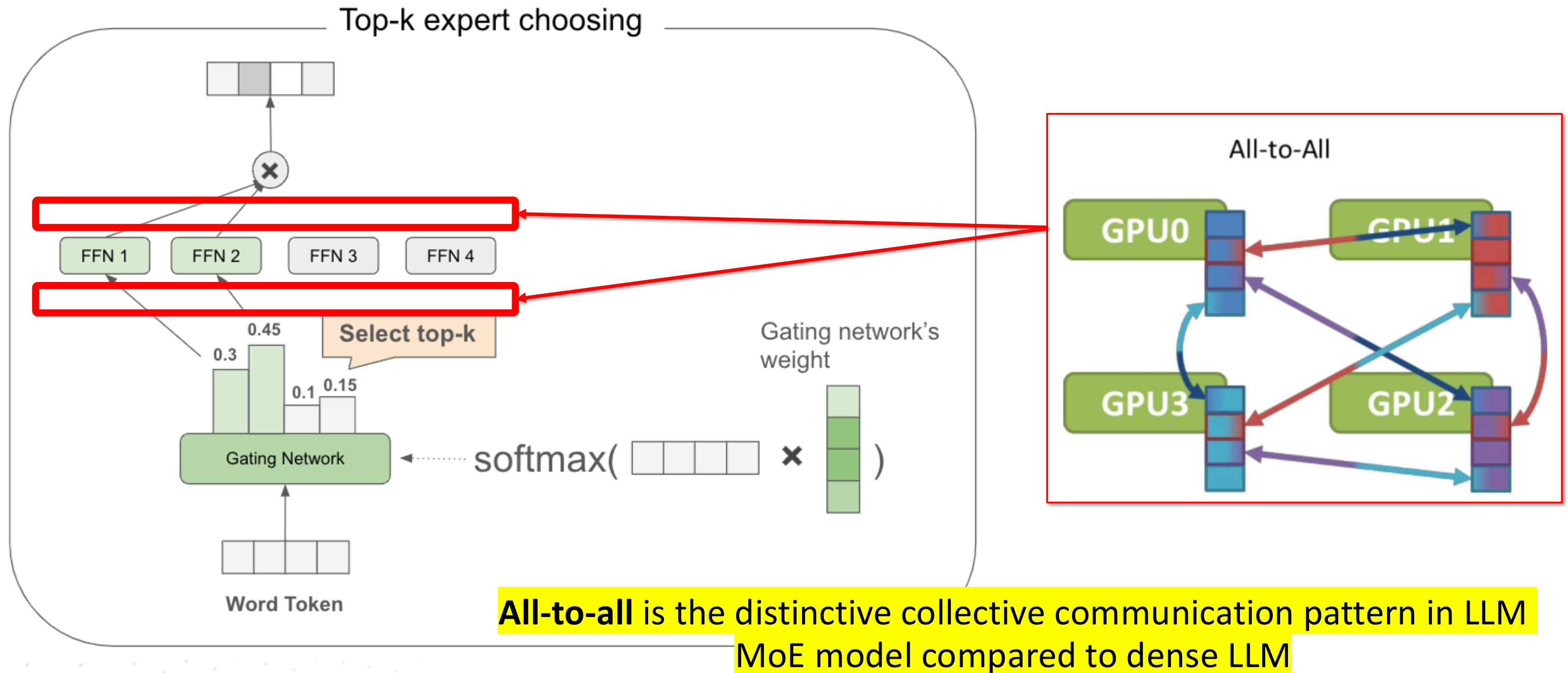


Models and datasets must be sub-divided to fit into GPU memory for distributed training

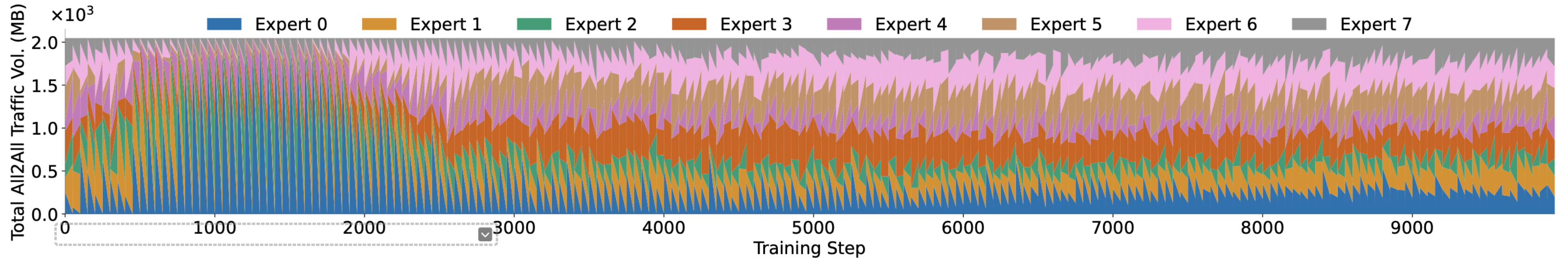
LLM Mixture-of-Experts (MoE)



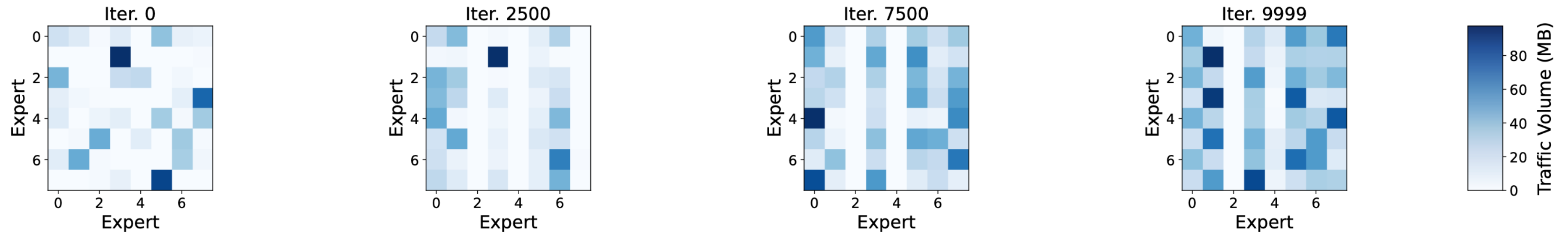
All-to-All traffic pattern in LLM MoE model



All-to-All traffic pattern in LLM MoE model



(a) In the temporal dimension, the all-to-all traffic volume of each node varies across different training iterations.



(b) In spatial dimension, the all-to-all traffic volume is non-uniform across different experts.

Figure 4: [Mixtral 8×7B in production] All-to-all traffic dynamics during MoE training.

Expert selection shows strong spatial locality which makes a **hotspot GPU**

Distributed Infrastructure for ML

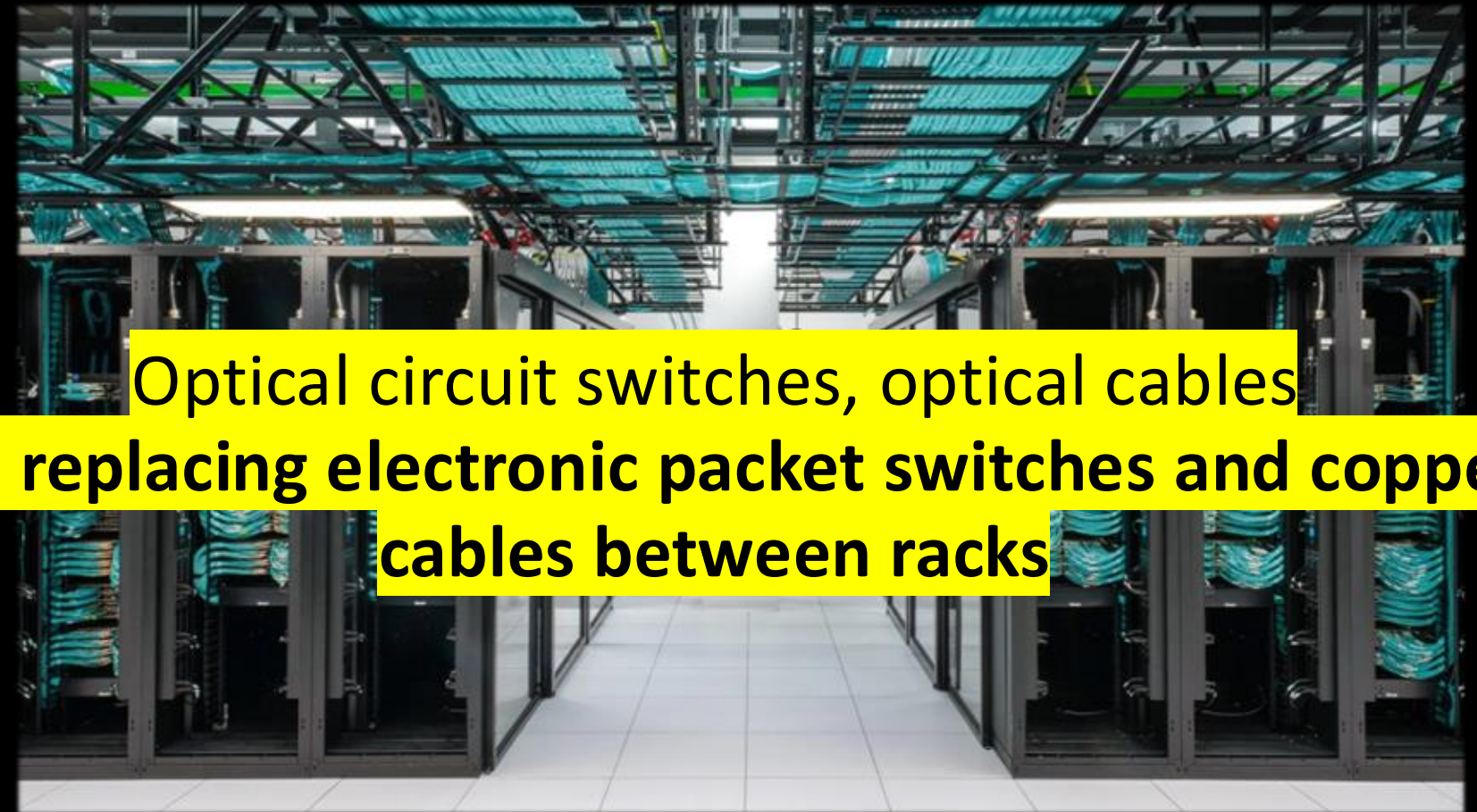
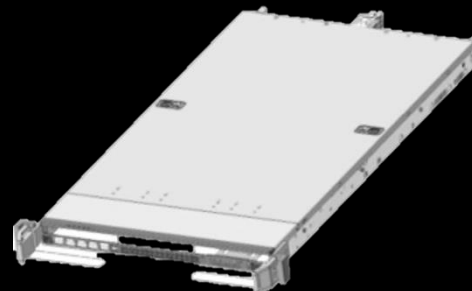
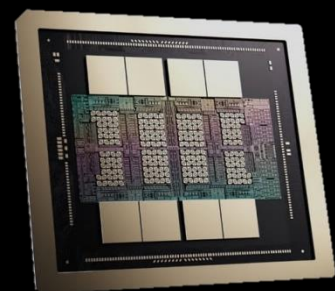
ML is changing datacenters, but how?

- Under-the-radar Datacenter Infrastructure
- Communication Hierarchy

Distributed Infrastructure for ML

How to introduce optical connections
between chips/trays?

Optical circuit switches, optical cables
are replacing electronic packet switches and copper
cables between racks



Chip

Tray

Rack

Data Center

Package

Between Trays

Between Racks

Between Data Center

Scale-Up

Scale-Out

Scale-Across

Communication Hierarchy

Taxonomy of Interconnects for AI Accelerators in Datacenter

Scale-Across

Between
Datacenters

How to introduce optical connections
between chips/trays?

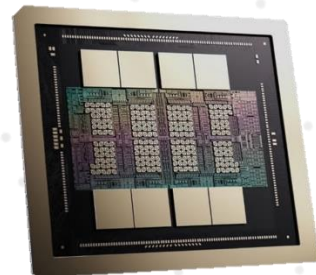
Scale-Up Domain

NVLink, UALink

Between GPUs and Switches

Inside a GPU

Package



Scale-Out Network

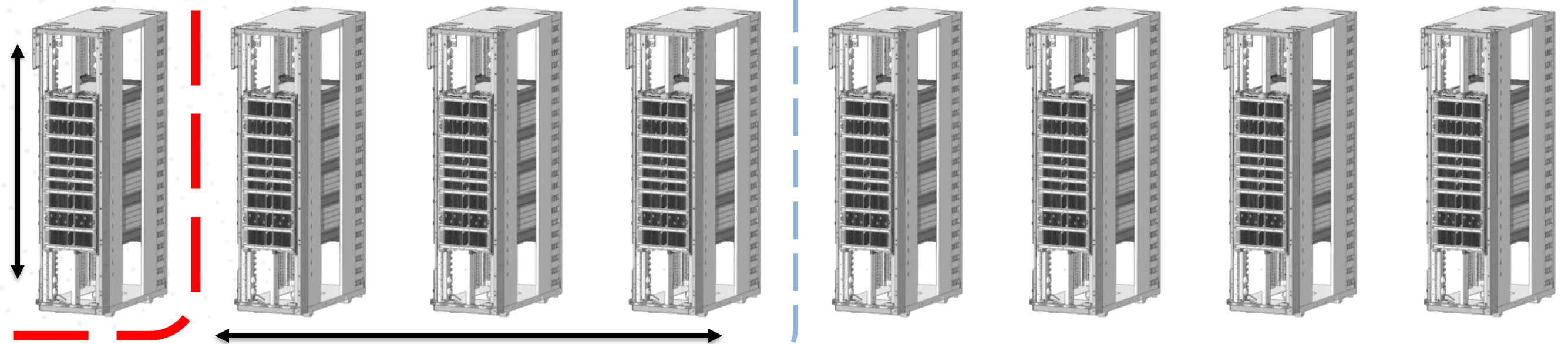
Ethernet or InfiniBand™

Connects accelerators in
AI Cluster

Front Side Network

Ethernet

Connects everything in the datacenter
(storage, compute tiers)

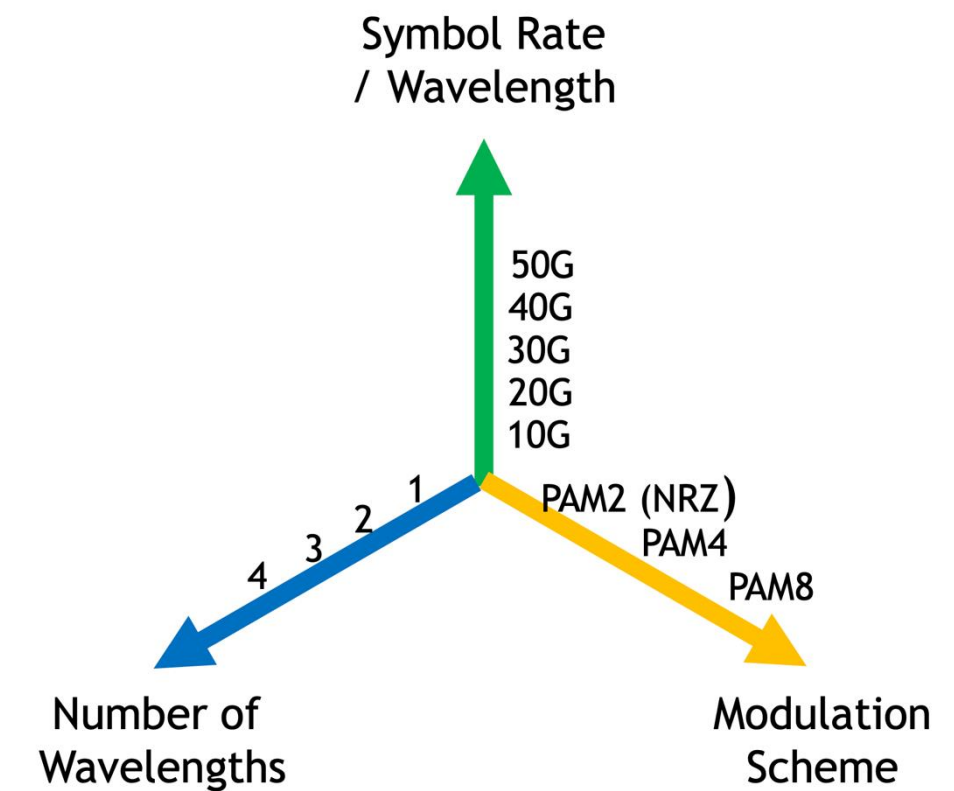
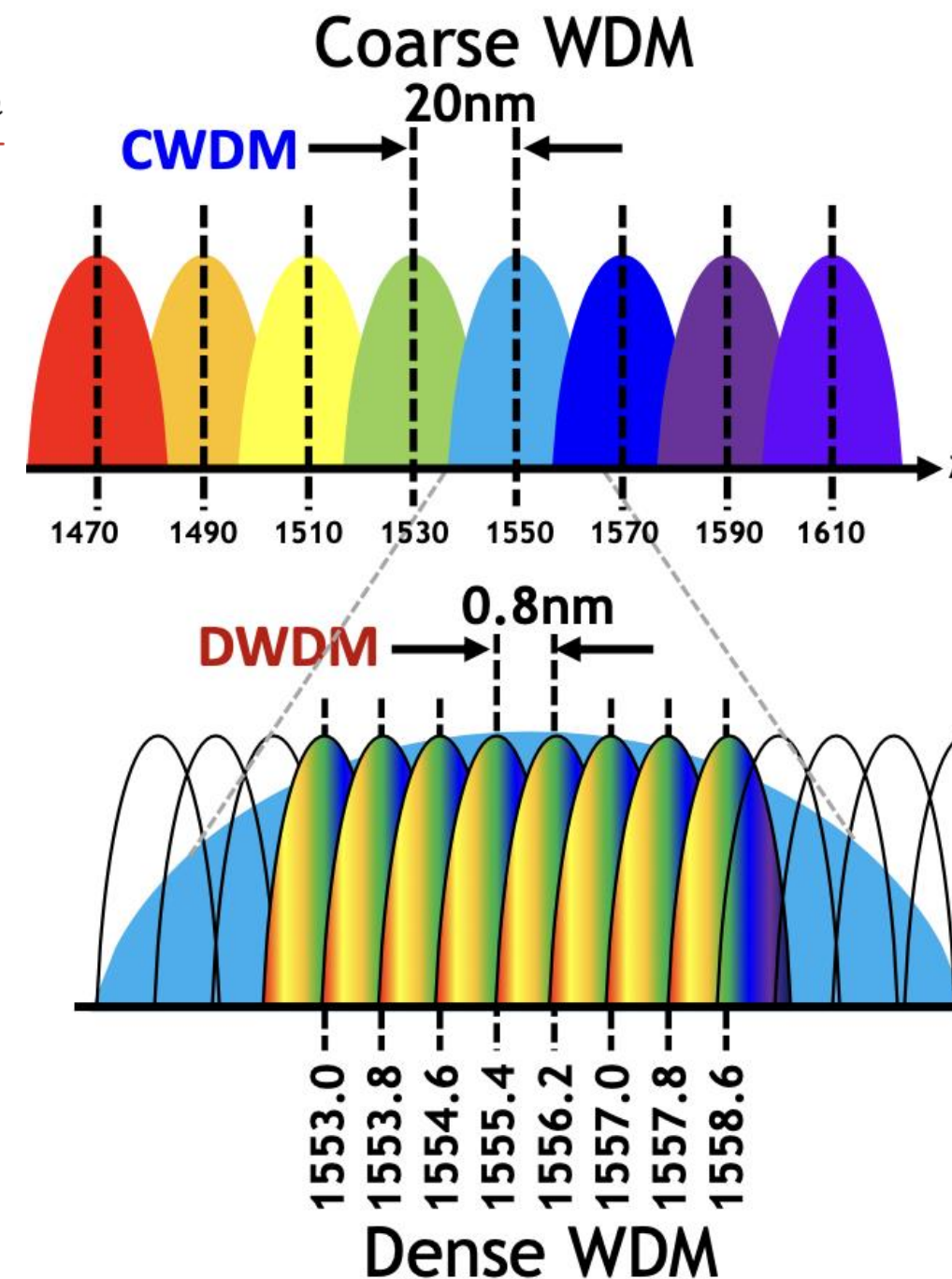
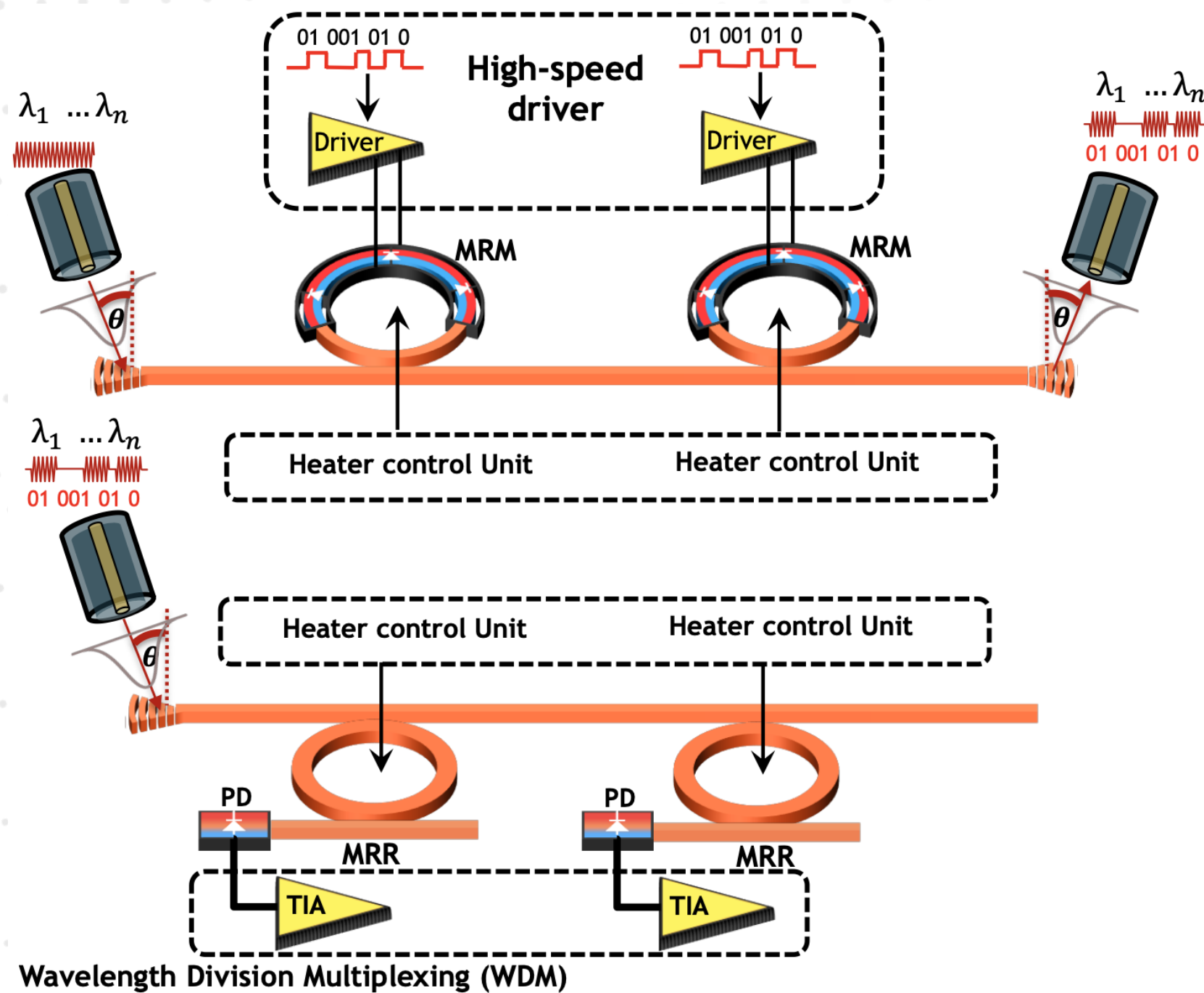


Optical Interconnect Solution in Scale-up Domain

How to introduce optical module?

- Optical Device Module
- 1. Co-Packaged Optics (CPO)
- 2. Optical Interposer

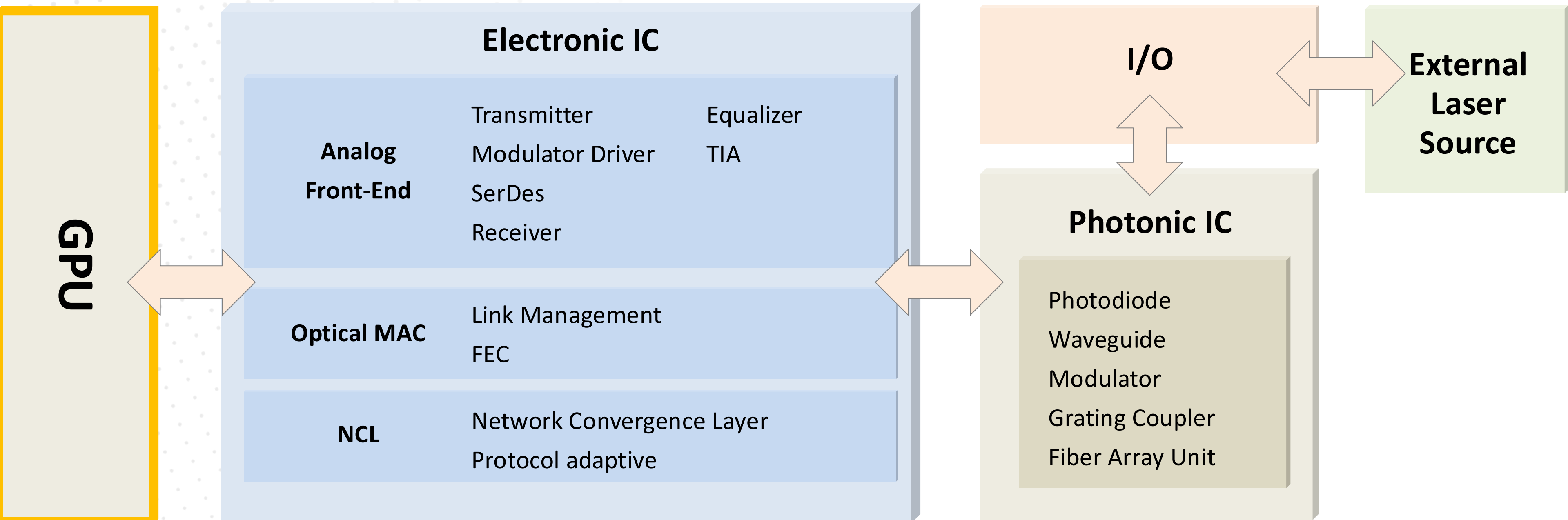
Bandwidth Scaling with Optical Interconnect



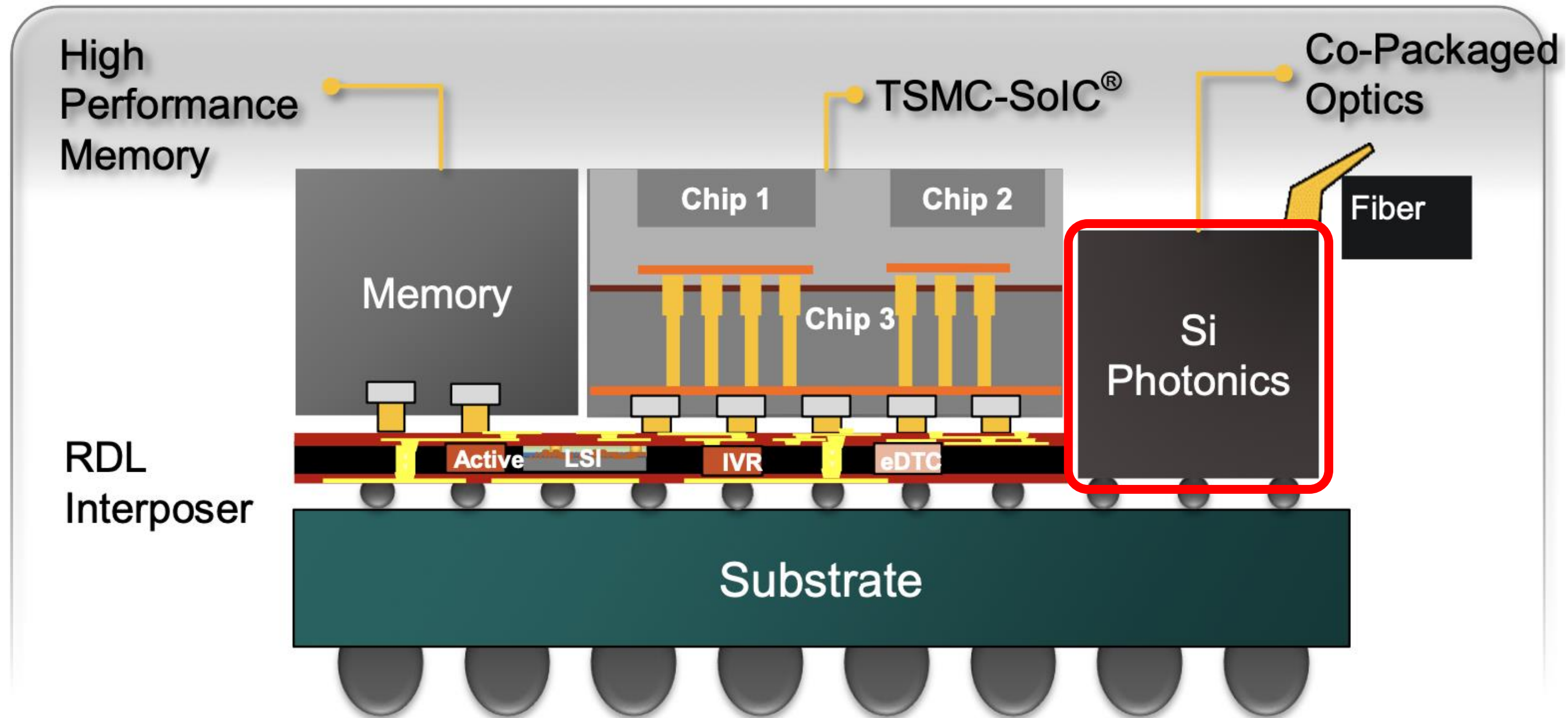
Wavelength Division Multiplexing enables bandwidth scaling

Optical Interconnect Overview

Optical connection requires electrical-optical conversion



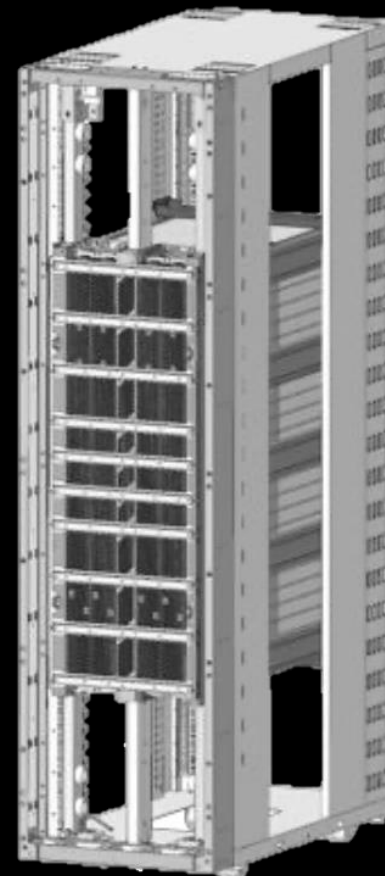
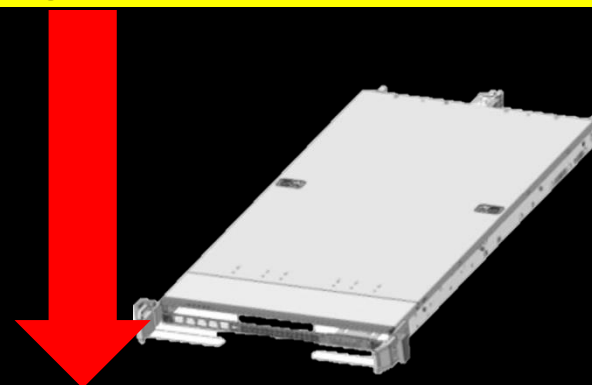
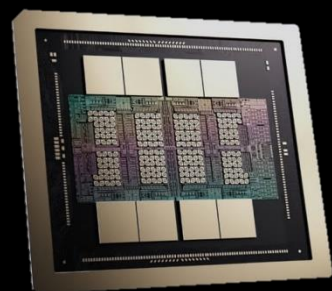
TSMC COUPE platform for optical connection



TSMC offers optical packaging platform with stacked EIC / PIC

Distributed Infrastructure for ML

Let's look at the two ways to introduce optical interconnect



Chip

Tray

Rack

Data Center

Package

Between Trays

Between Racks

Between Data Center

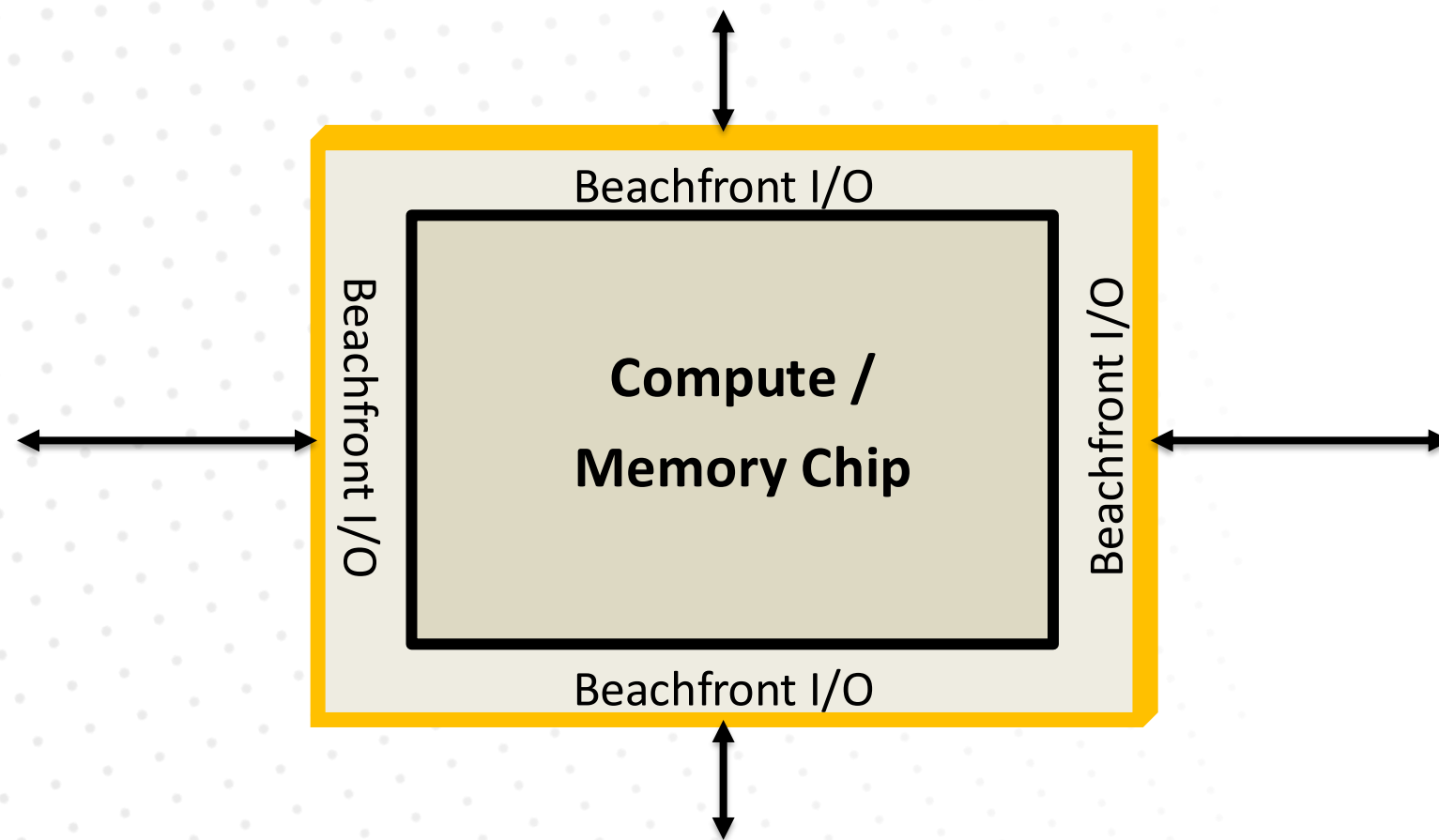
Scale-Up

Scale-Out

Scale-Across

How to introduce Optical Module in Scale-up Domain?

Two options



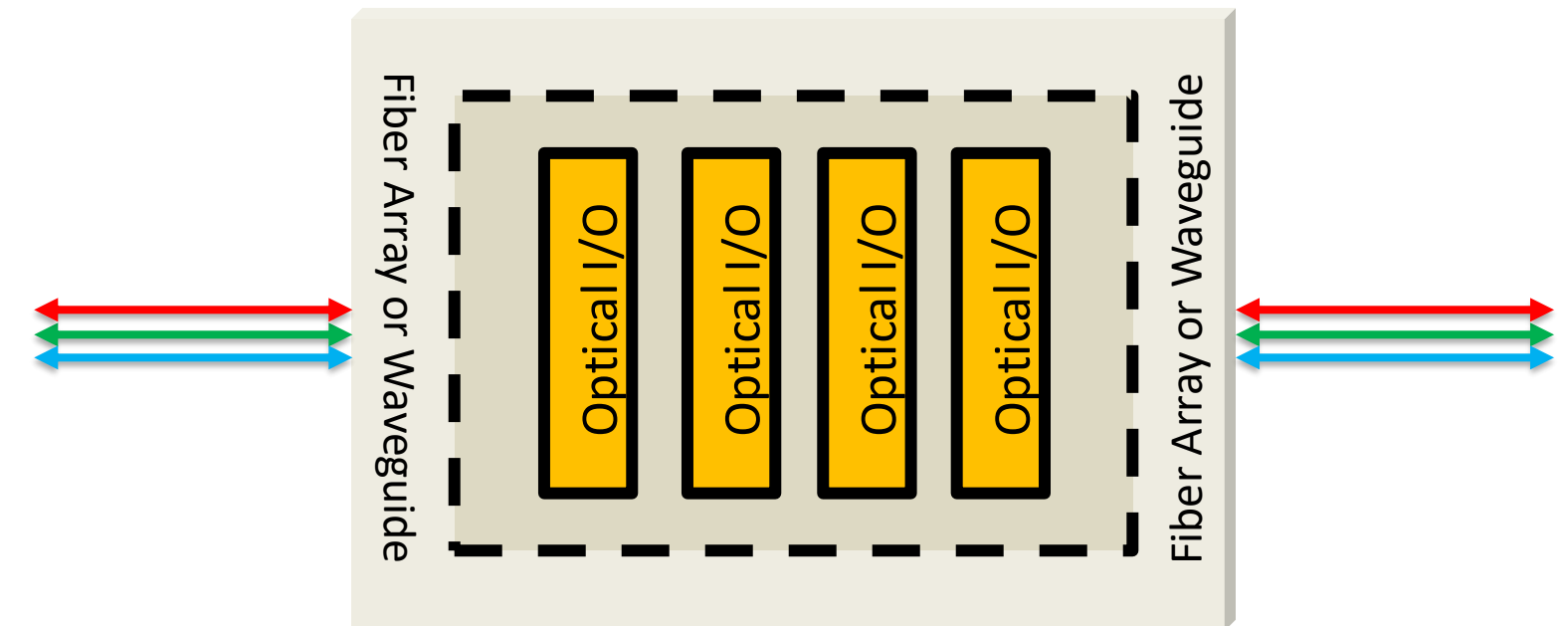
1. Electronic Fabric + CPO

PCIe, UCIe

Communication happens at the chip parameter

$\text{FLOPs} \propto \text{Chip Area } (L^2)$

$\text{I/O Bandwidth} \propto \text{Chip Perimeter } (L)$



2. Optical Interposer

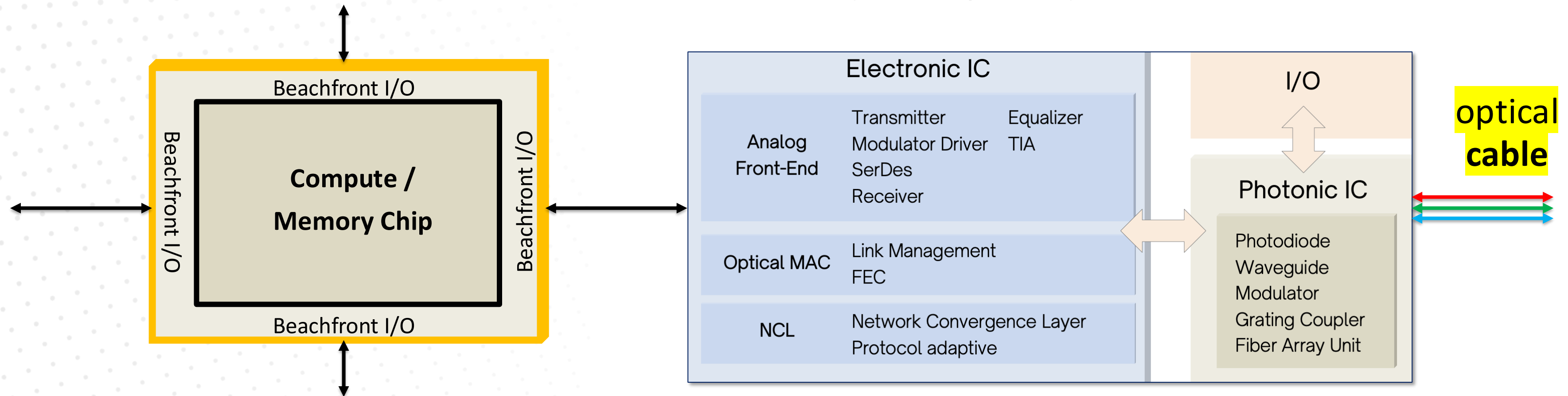
Silicon Photonics

I/O Bandwidth scales with compute area
by 3D packaging

$\text{I/O Bandwidth} \propto \text{Chip Area } (L^2)$

How to introduce Optical Module in Scale-up Domain?

1. Electronic fabric + co-packaged optics



1. Electronic Fabric + CPO

PCIe, UCIe

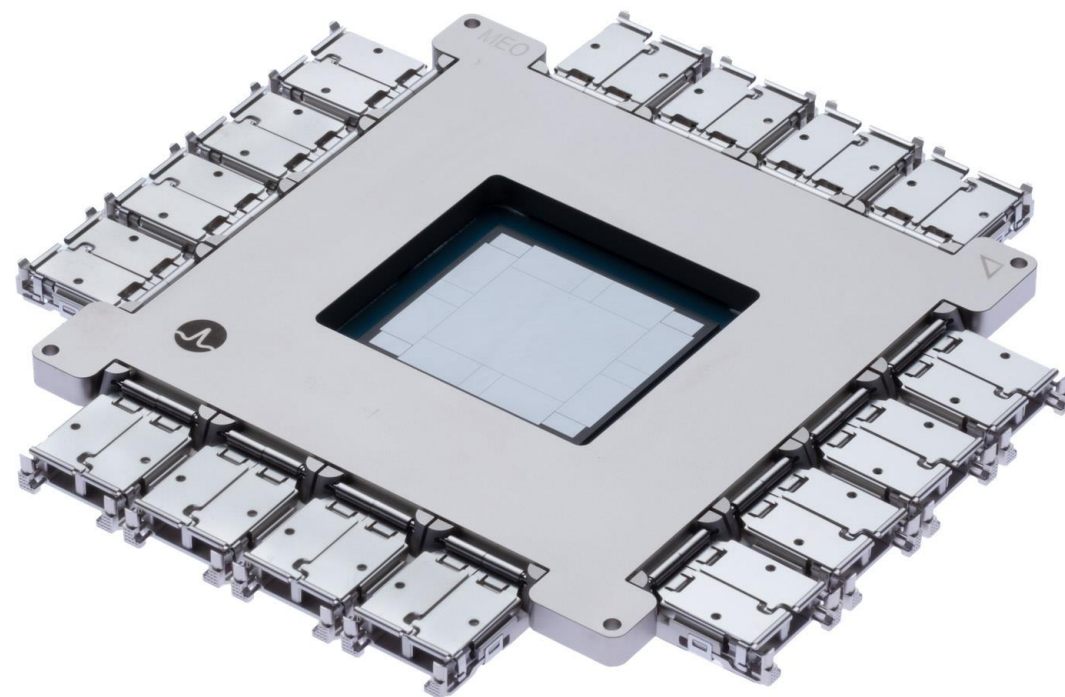
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$$\text{FLOPs} \propto \text{Chip Area } (L^2)$$

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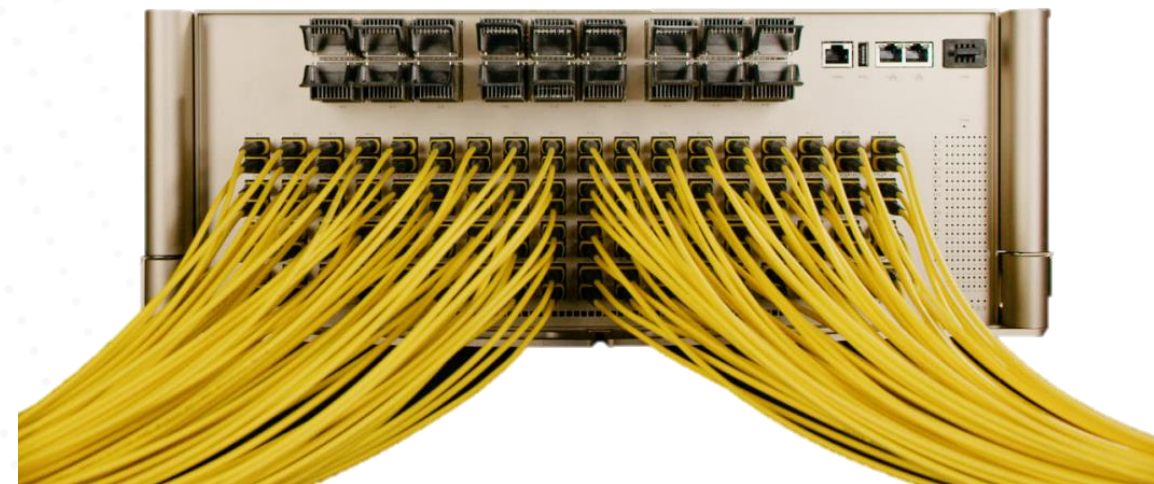
First option :
conventional electronic fabric with
co-packaged optics

Co-Packaged Optics (CPO) in 2.5D Integration



Tomahawk

Ethernet switch ASIC
102.4 Tbps



Quantum-X Photonics Switch

InfiniBand switch system / platform
144 ports x 800 Gb/s (115.2 Tbps)

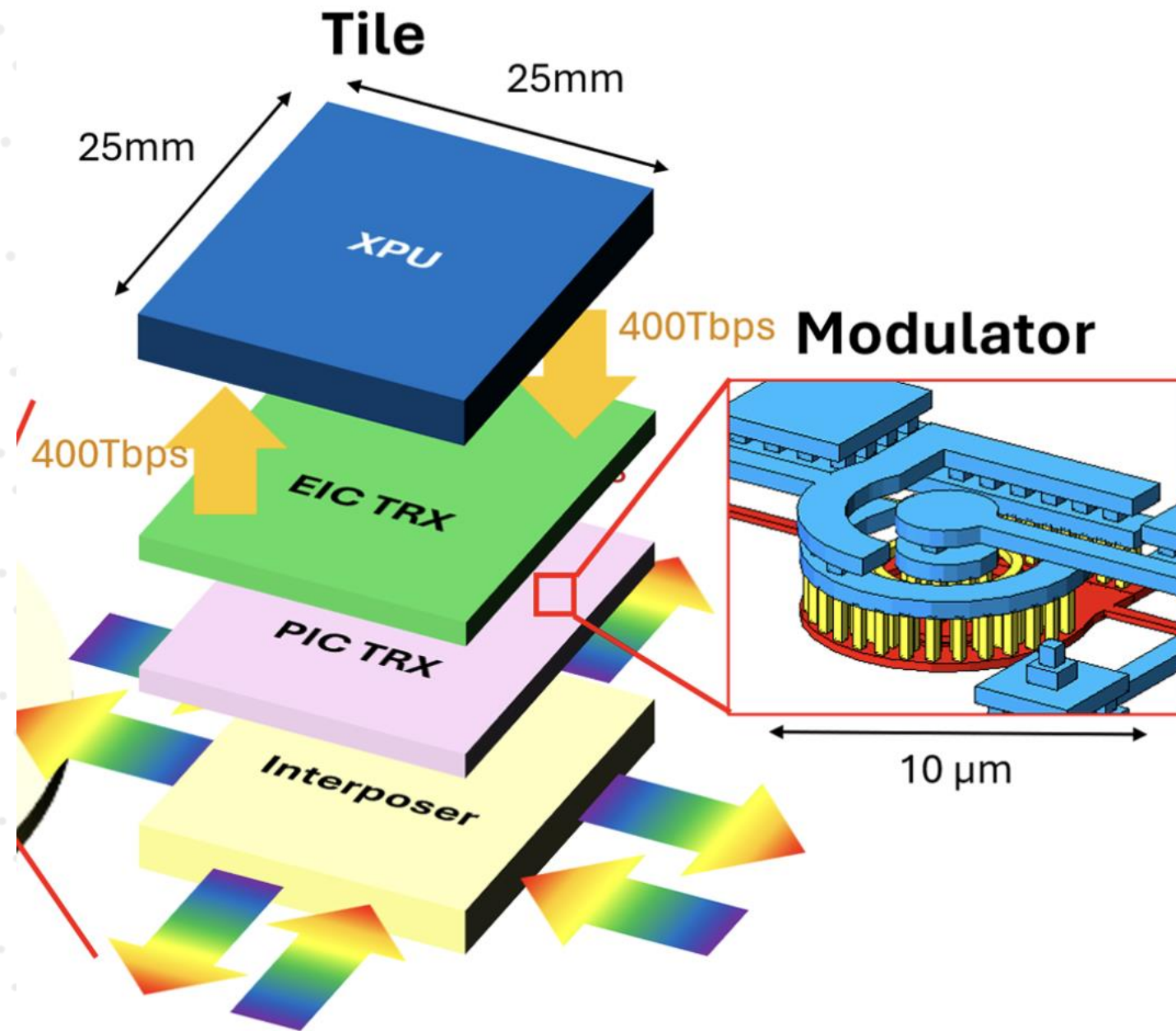


TeraPHY™ Optical Engine

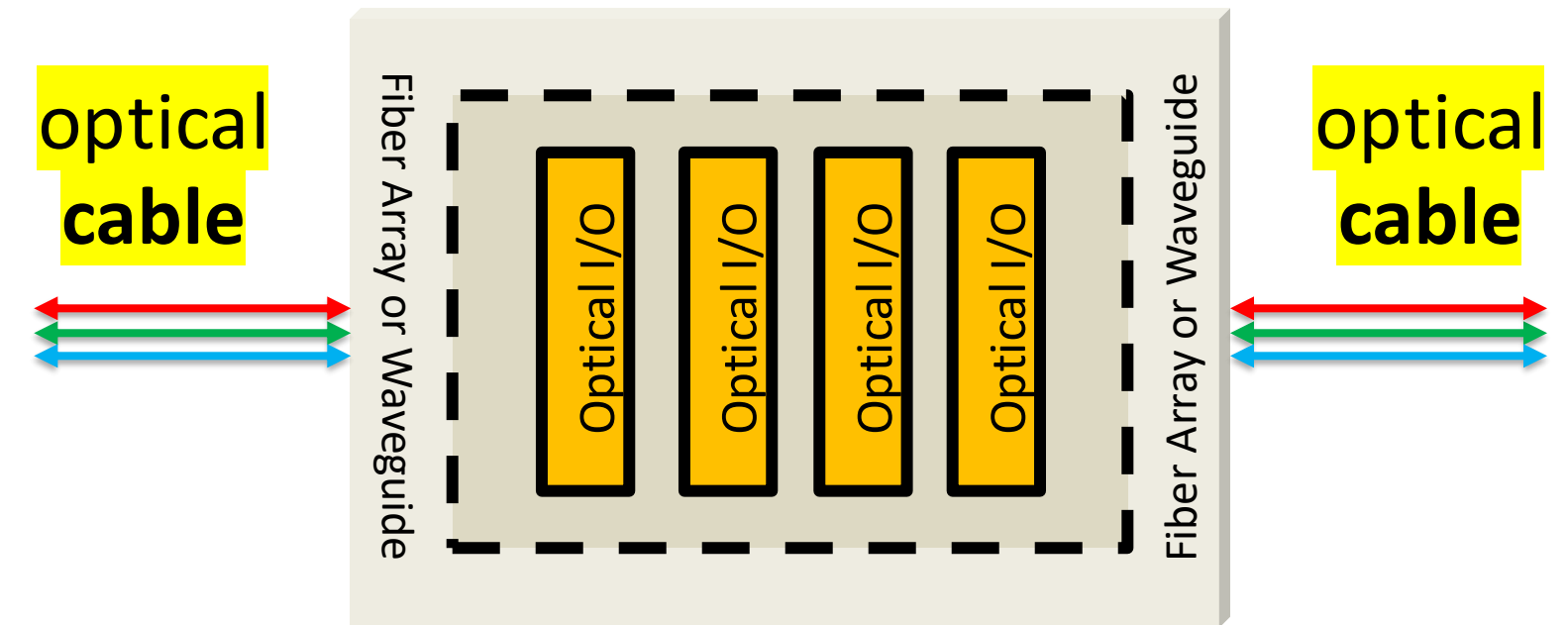
UCle-based optical I/O chiplets
8 Tbps per chiplet

How to introduce Optical Module in Scale-up Domain?

2. Optical Interposer



Second option : waveguide based
GPU - EIC - PIC stack 3D Integration



2. Optical Interposer

Silicon Photonics

I/O Bandwidth scales with compute area
by 3D packaging

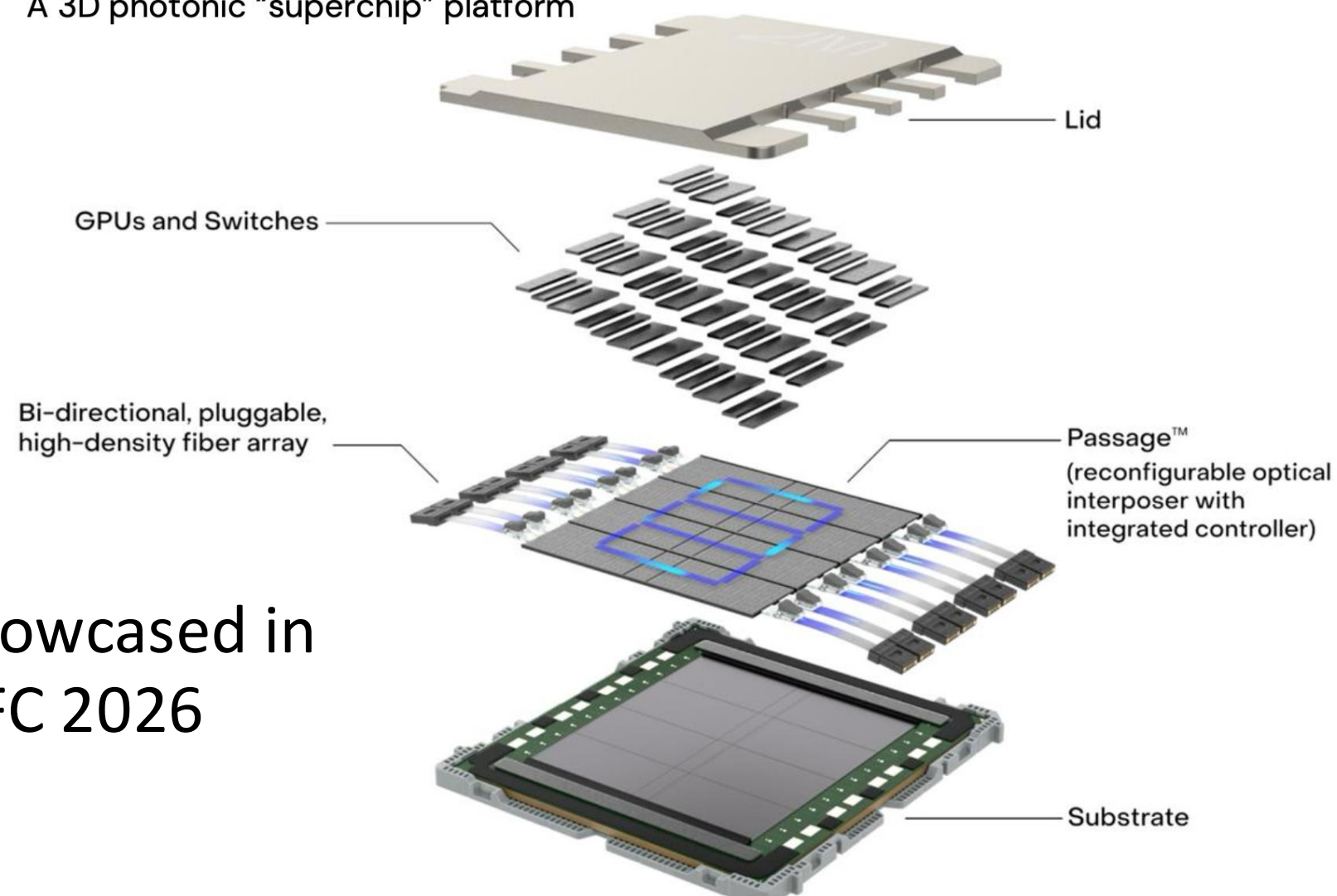
$$\text{I/O Bandwidth} \propto \text{Chip Area } (L^2)$$

Optical Interposer with 3D integration

LIGHTMATTER

Passage M1000

A 3D photonic “superchip” platform

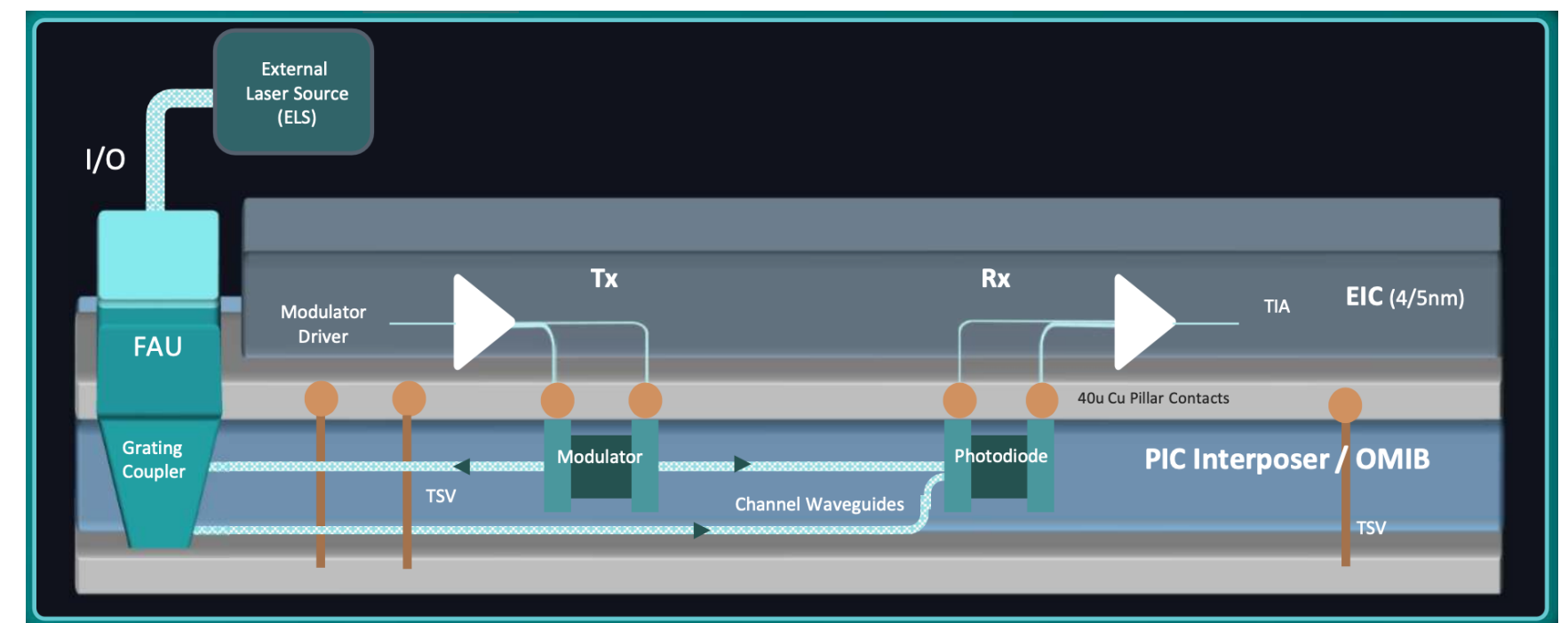


Showcased in
OFC 2026

Top Layer : Compute and memory chiplets

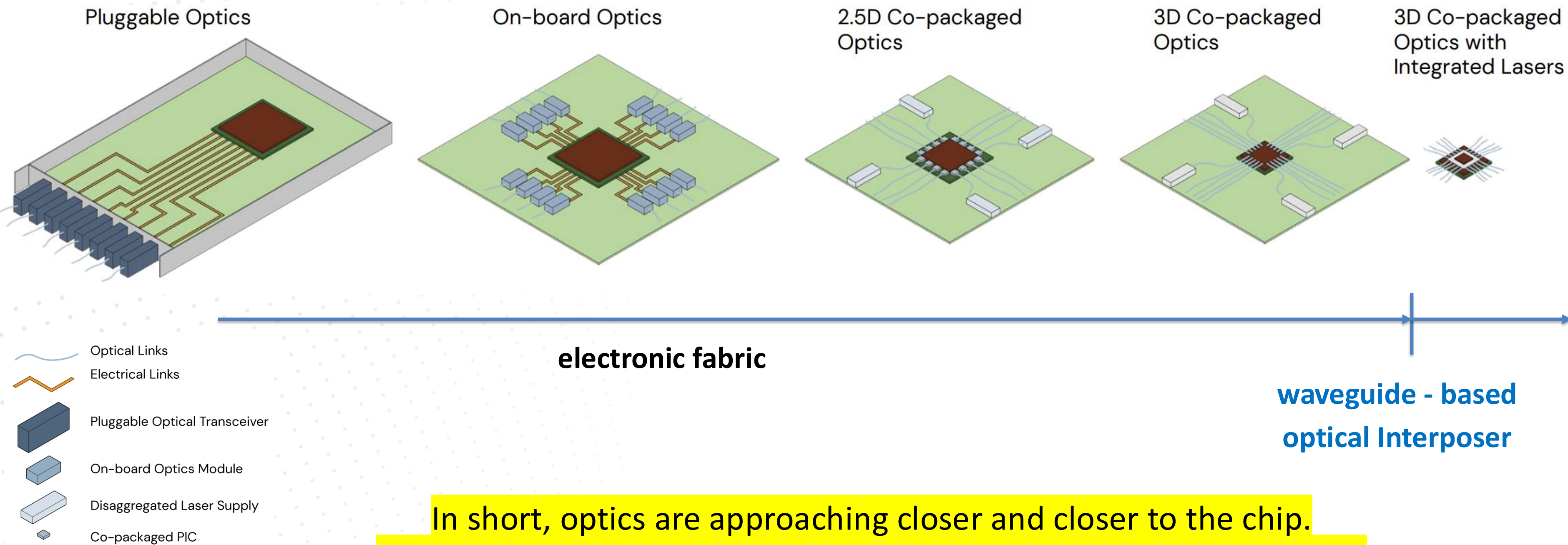
Base Layer : lasers, waveguides, and optical switching and routing

celestial AI™



Key advantage of optical interposer :
Delivers continuous 2D surface for optical I/O

Optical interconnect in chip- rack scale summary

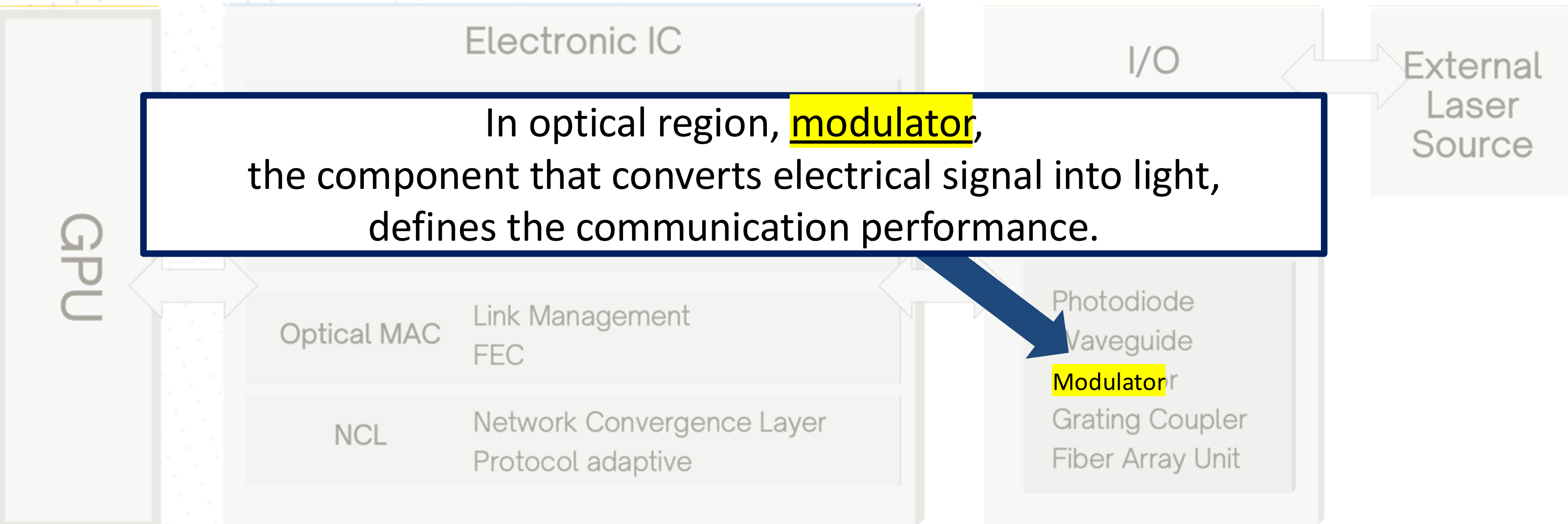


In short, optics are approaching closer and closer to the chip.
But what is the bottleneck in bringing optics closer to the processor?

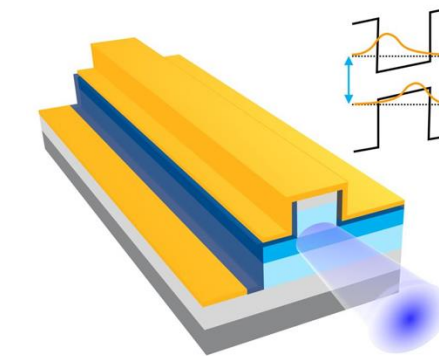
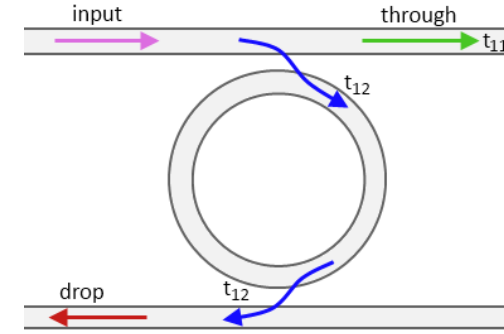
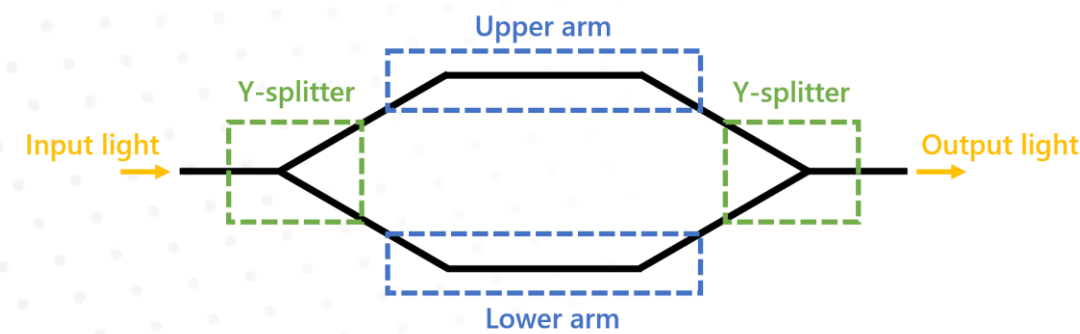
Recall Optical Interconnect System

Optical connection requires electrical-optical conversion

In optical region, modulator,
the component that converts electrical signal into light,
defines the communication performance.



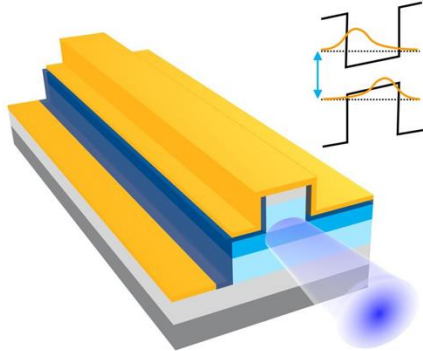
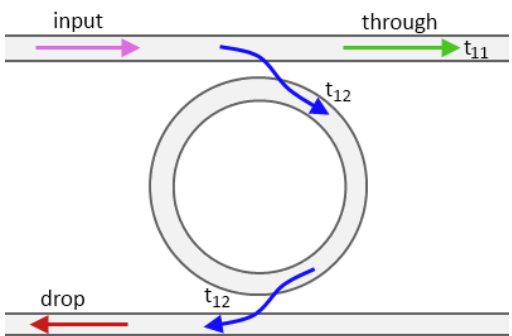
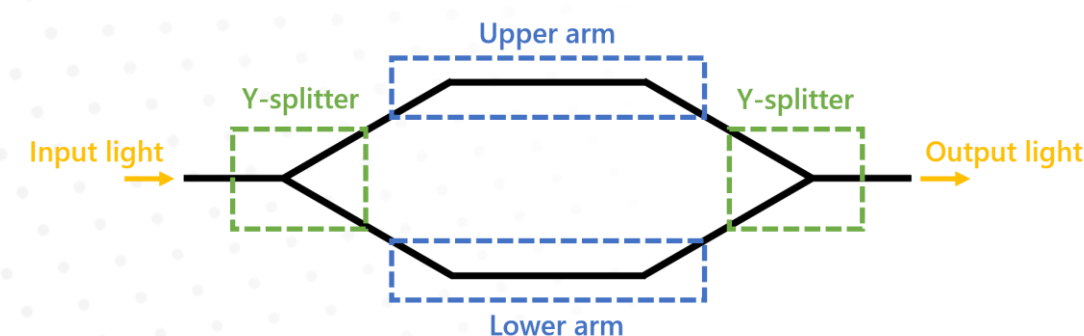
Modulators inside PICs are based on silicon photonics



Metric	Mach-Zehnder Interferometer (MZI)	Microring Resonator (MRR) <i>Mostly thermo-optic</i>	Electro-Absorption Modulator (EAM)
Size	Large (1000 - 2000 μm)	Very small (< 50 μm)	Small (30 - 100 μm)
Thermal Stability	Robust (Stable for over 50K fluctuation)	Sensitive (Must be stable under 1K range)	Good (Stable for over 50K fluctuation)
Speed (Data Rate)	< 56 Gbps	< 56 Gbps	< 56 Gbps
Power	High (require tens of voltages)	Low	Low
Optical Bandwidth	Broadband	< 1nm	$\sim 10\text{s nm}$
WDM-friendly	Low (require MUX)	High (resonance shift)	Low (require MUX)
Operation Complexity	Low	High (resonance shift)	Low
XPU Integration	Low (high power)	Low (thermal crosstalk)	High

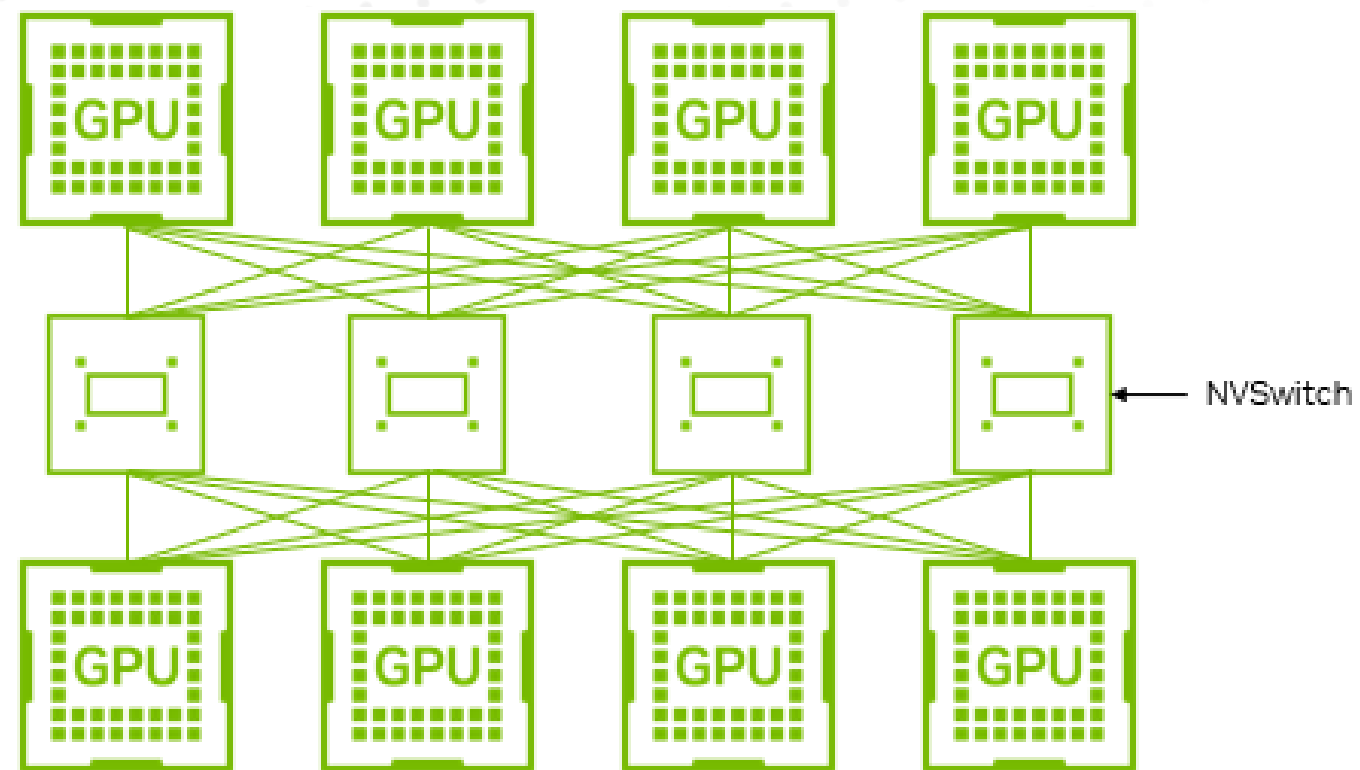
thermally-stable / electro-optic based

We propose **Ferroelectric** MRR

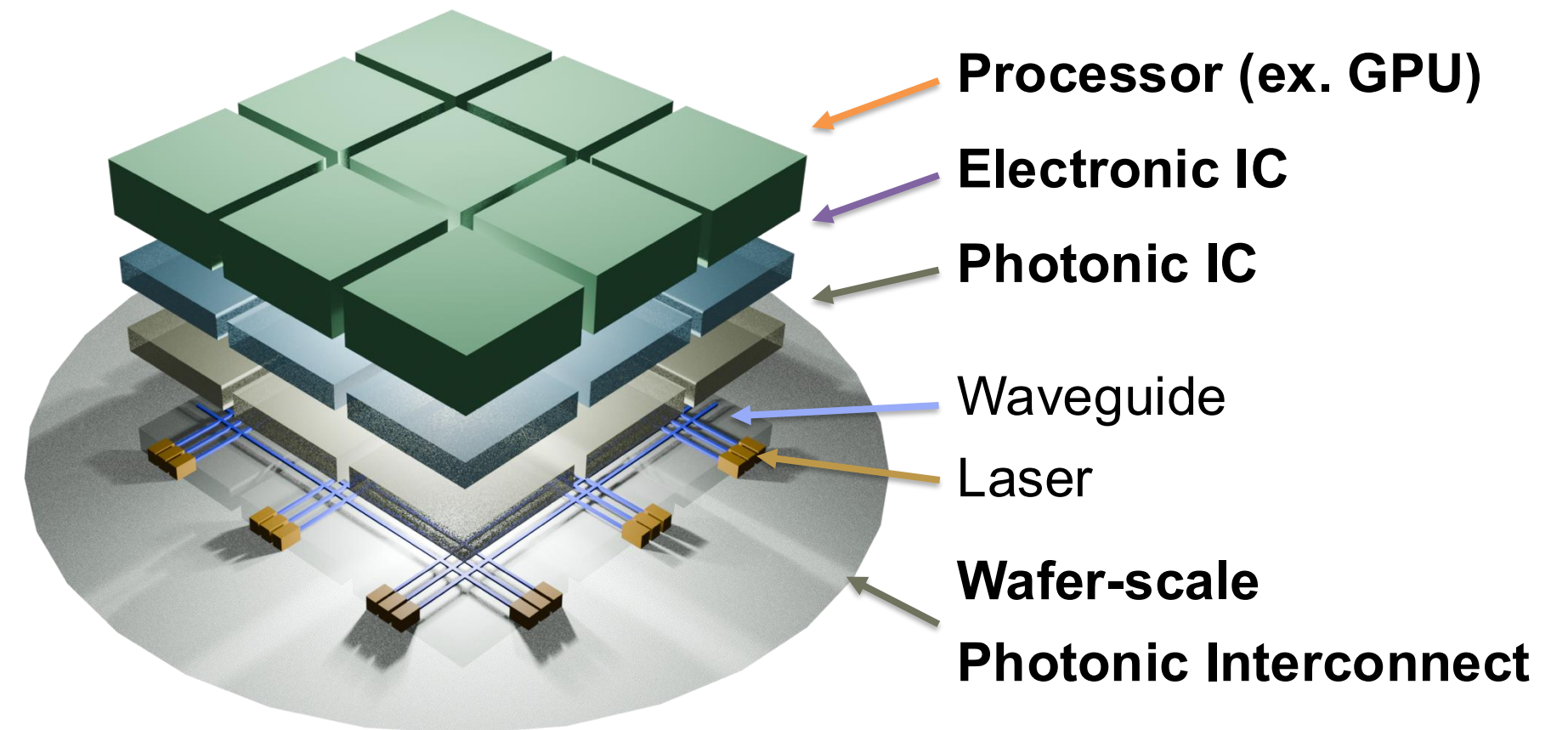


Metric	Mach-Zehnder Interferometer (MZI)	Microring Resonator (MRR) Mostly thermo-optic	Electro-Absorption Modulator (EAM)
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Operation Complexity	Low	High (resonance shift) Low	Low
XPU Integration	Low (high power)	Low (thermal crosstalk) High	High

Wafer/Panel-Scale Computing Platform



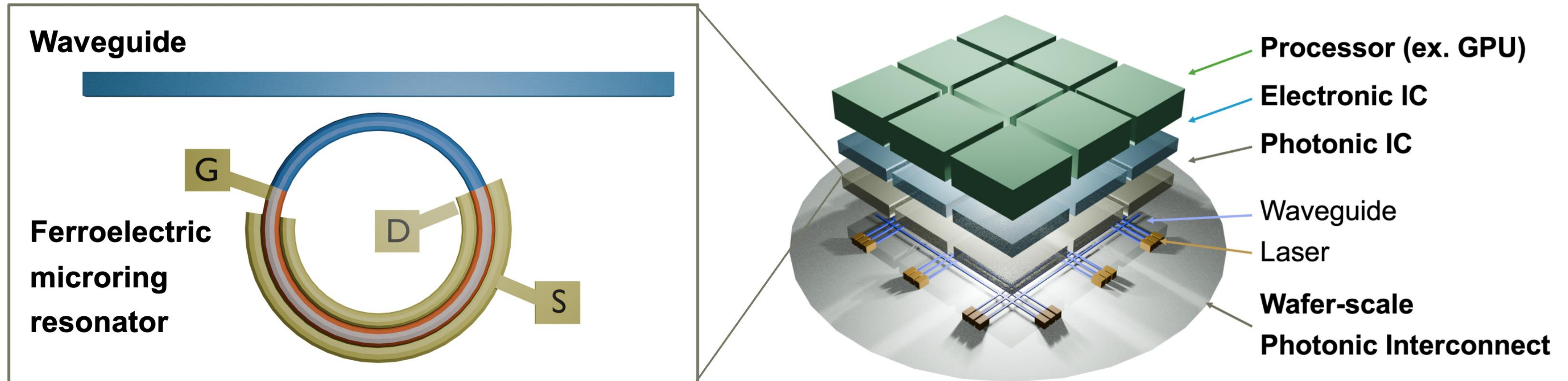
Nvidia DGX



Wafer-scale Platform

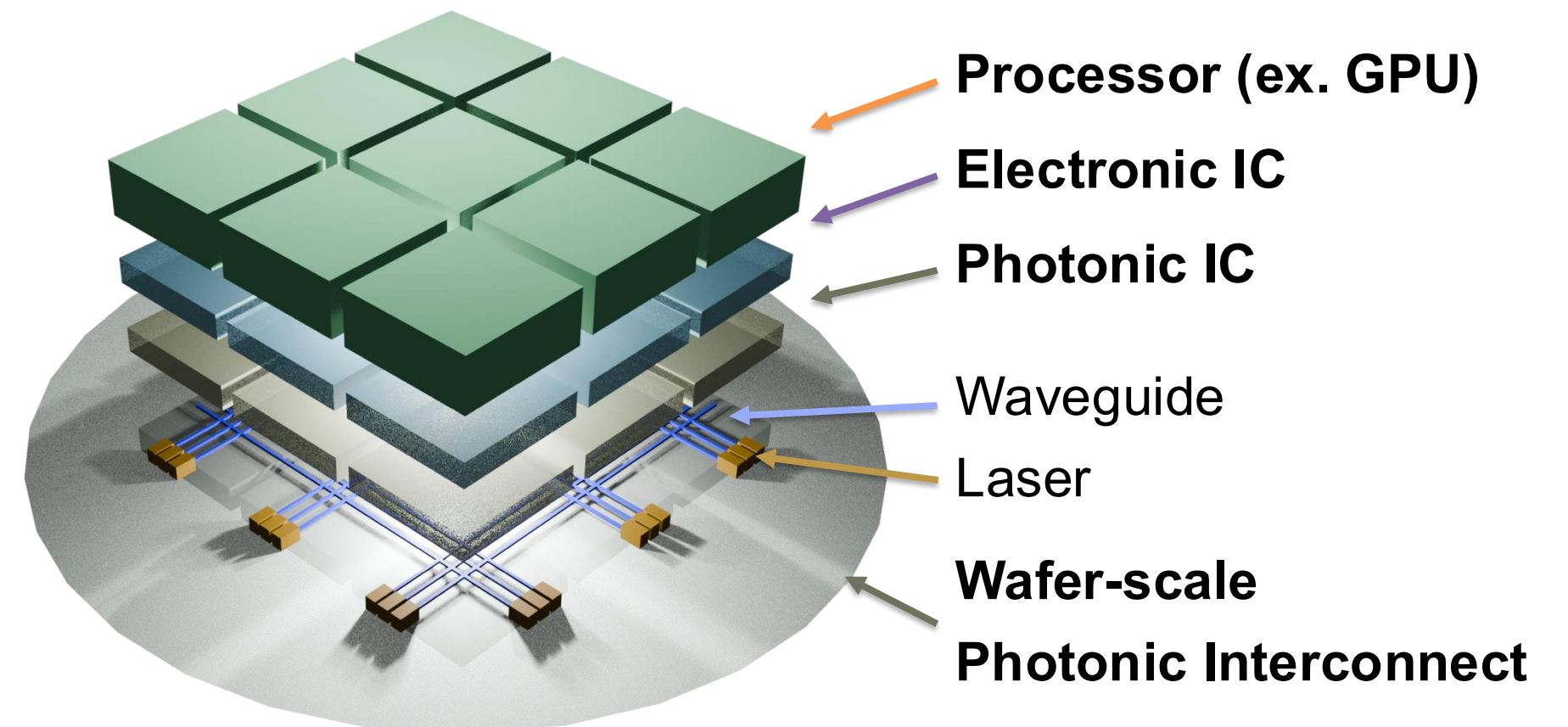
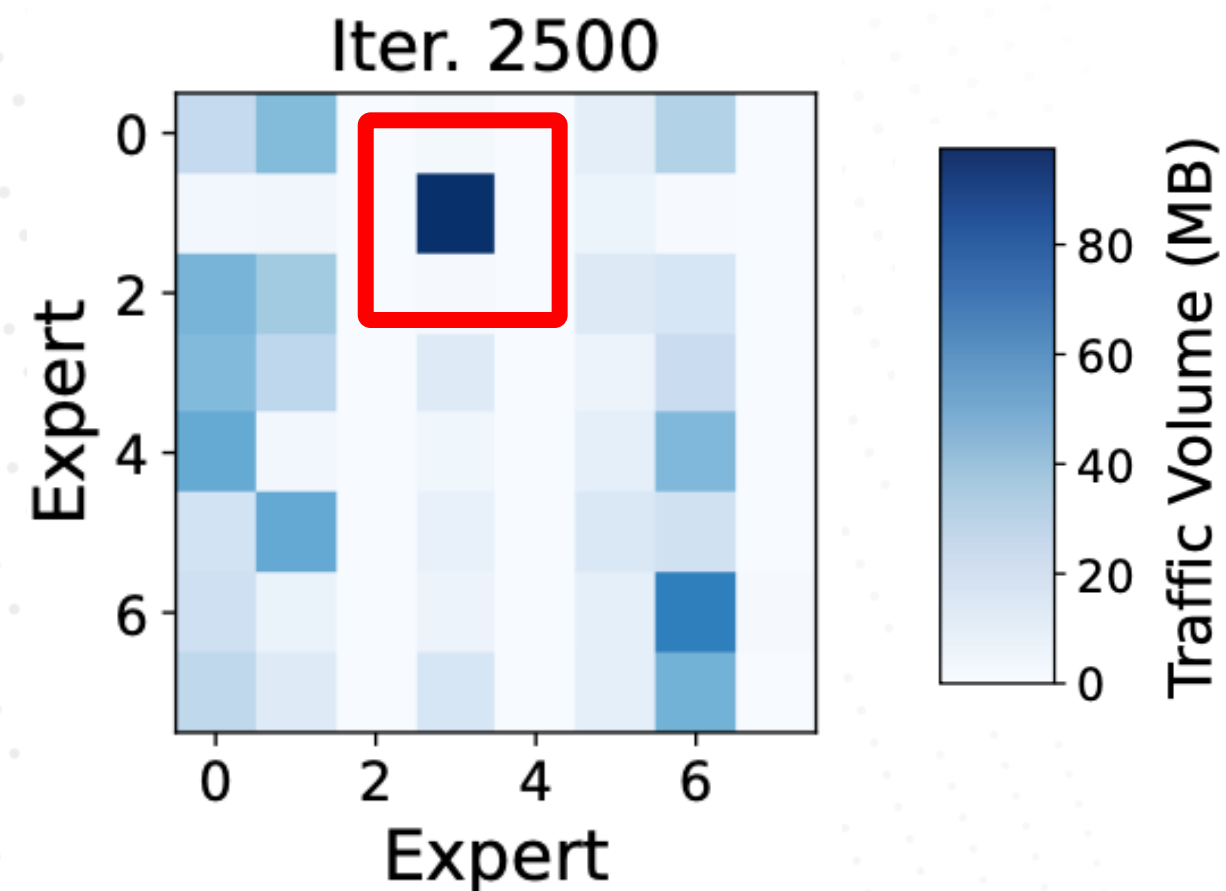
Wafer/panel-scale platform significantly increases the inter-GPU bandwidth compared to conventional multi-GPU node system.

Ferroelectric MRR in wafer-scale optical interposer



However, thermal MRRs in the underlying PIC is prone to thermal fluctuation.

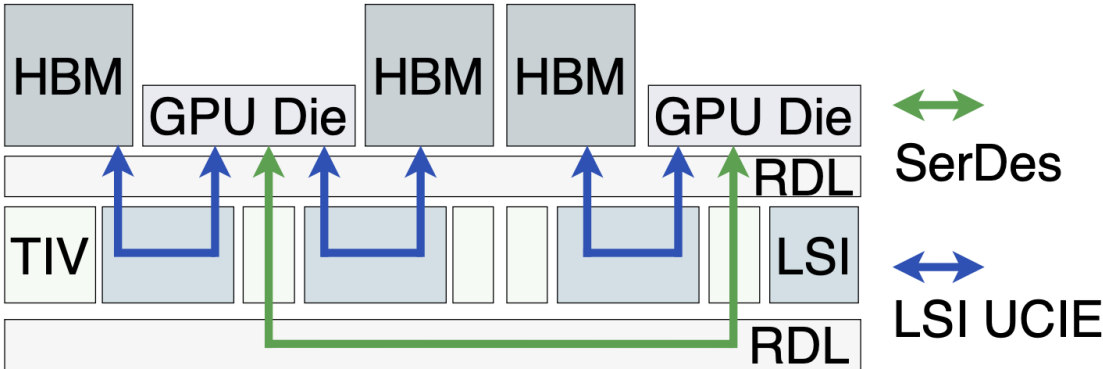
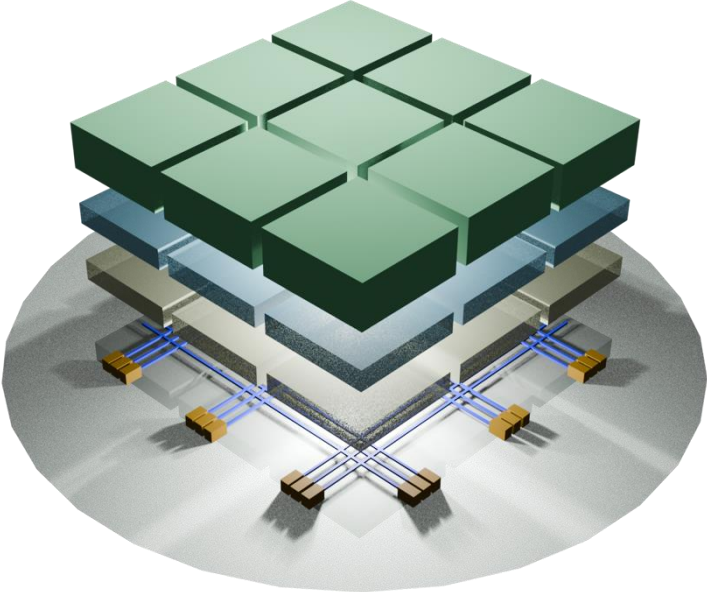
Ferroelectric MRR in wafer-scale optical interposer



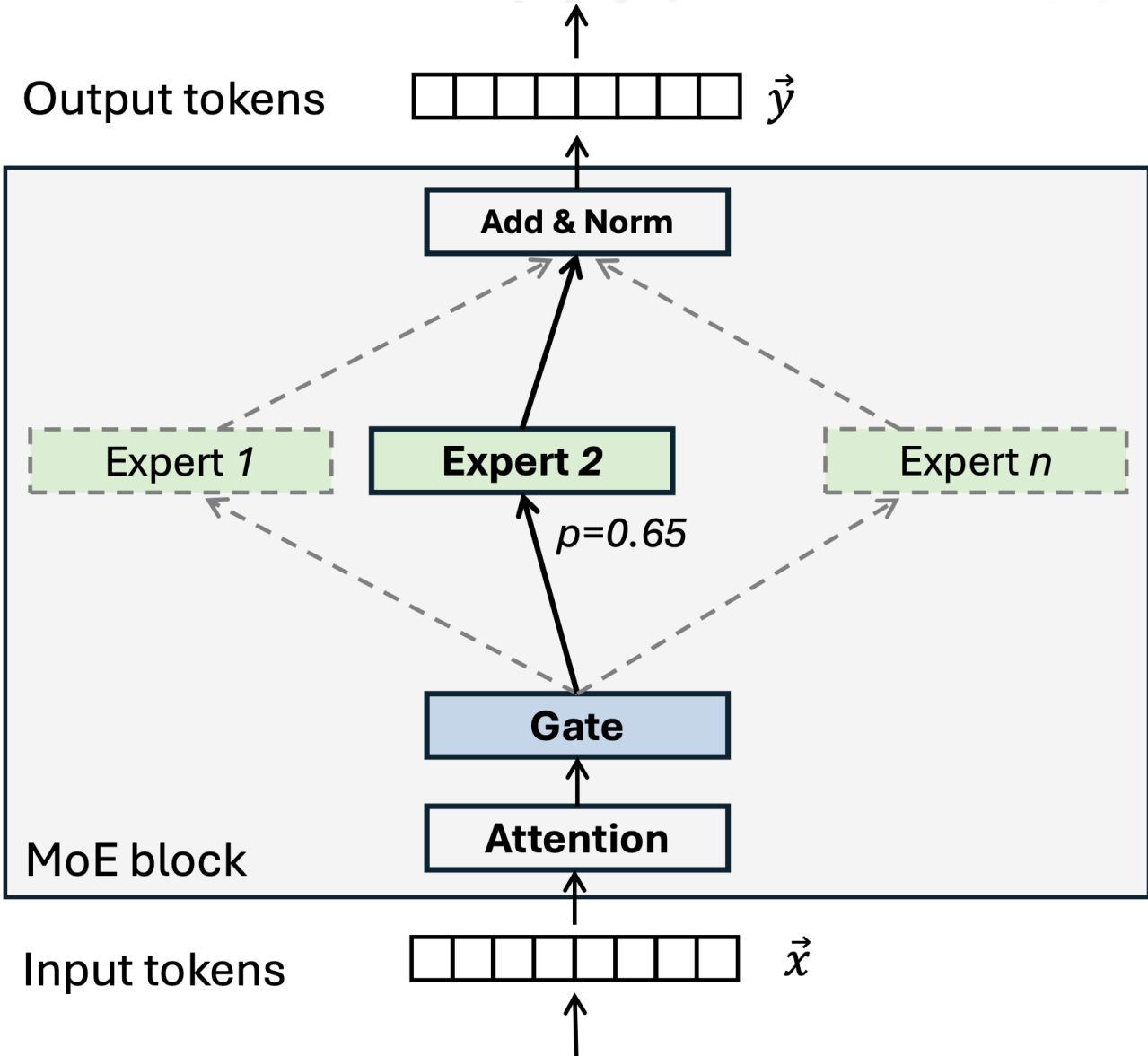
However, thermal MRRs in the underlying PIC is prone to thermal fluctuation.

We introduce ferroelectric MRRs simulated in LLM MoE model training scenario in which thermal hotspots are generated in **experts computation**

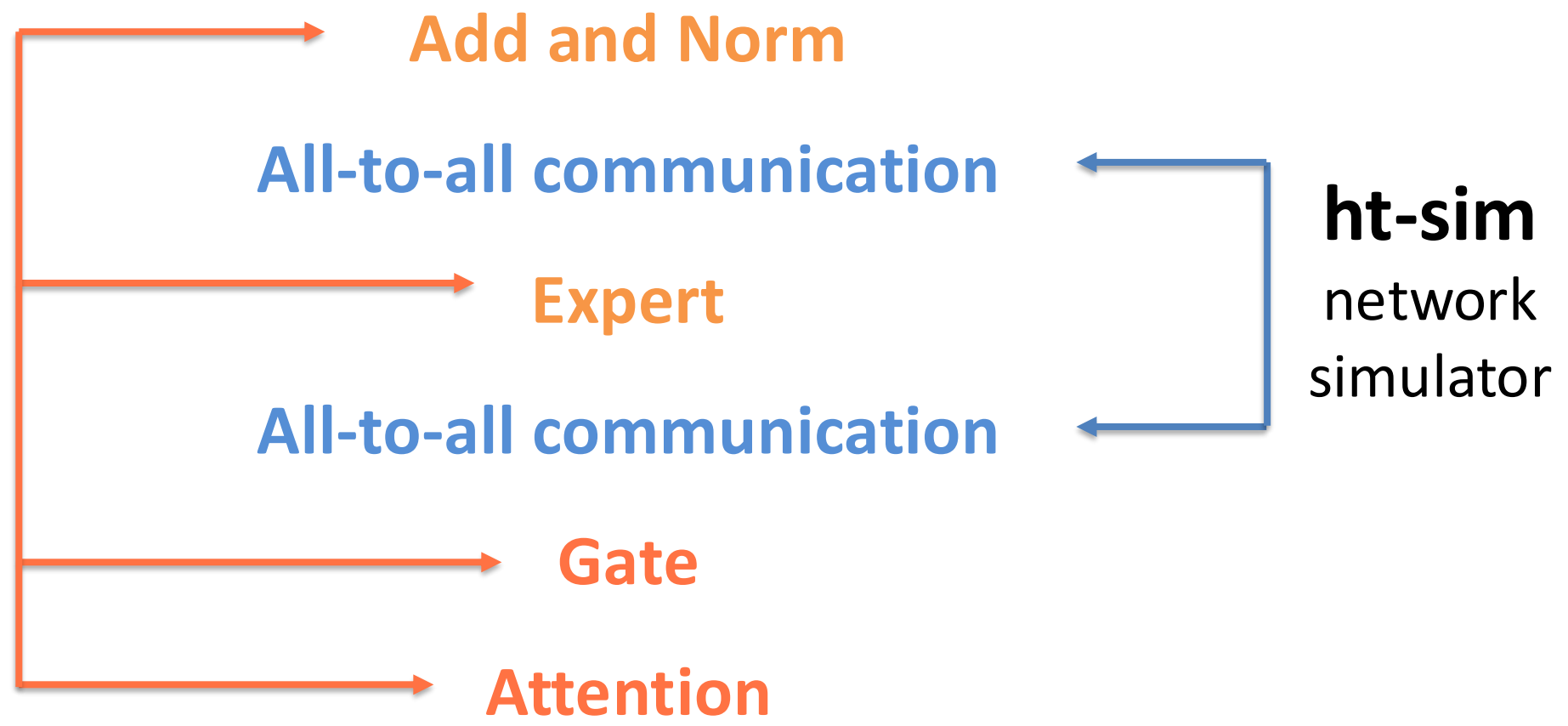
System-on-Wafer Architecture Comparison

	GPU	D2D Bandwidth	Hops	Latency	Structure
System-on-Wafer (electronic fabric)	4 x 4	1.5 TB/s	max 6	50 ns	
System-on-Wafer with optical interposer	4 x 4	1.5 TB/s	1 or 2	39 ns	 Thermal MRR Ferroelectric MRR

Profile LLM MoE models



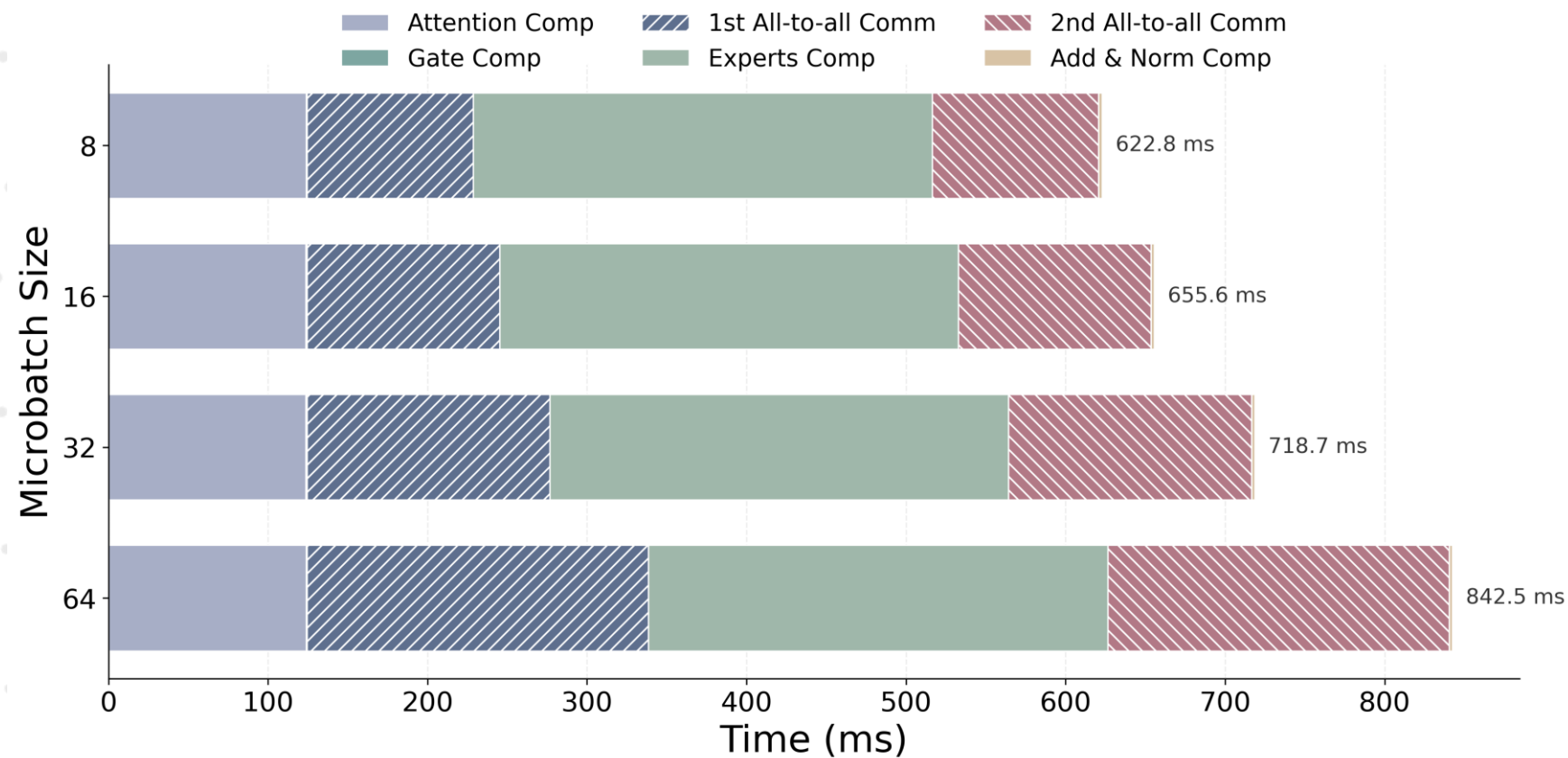
FlexFlow
DNN simulator



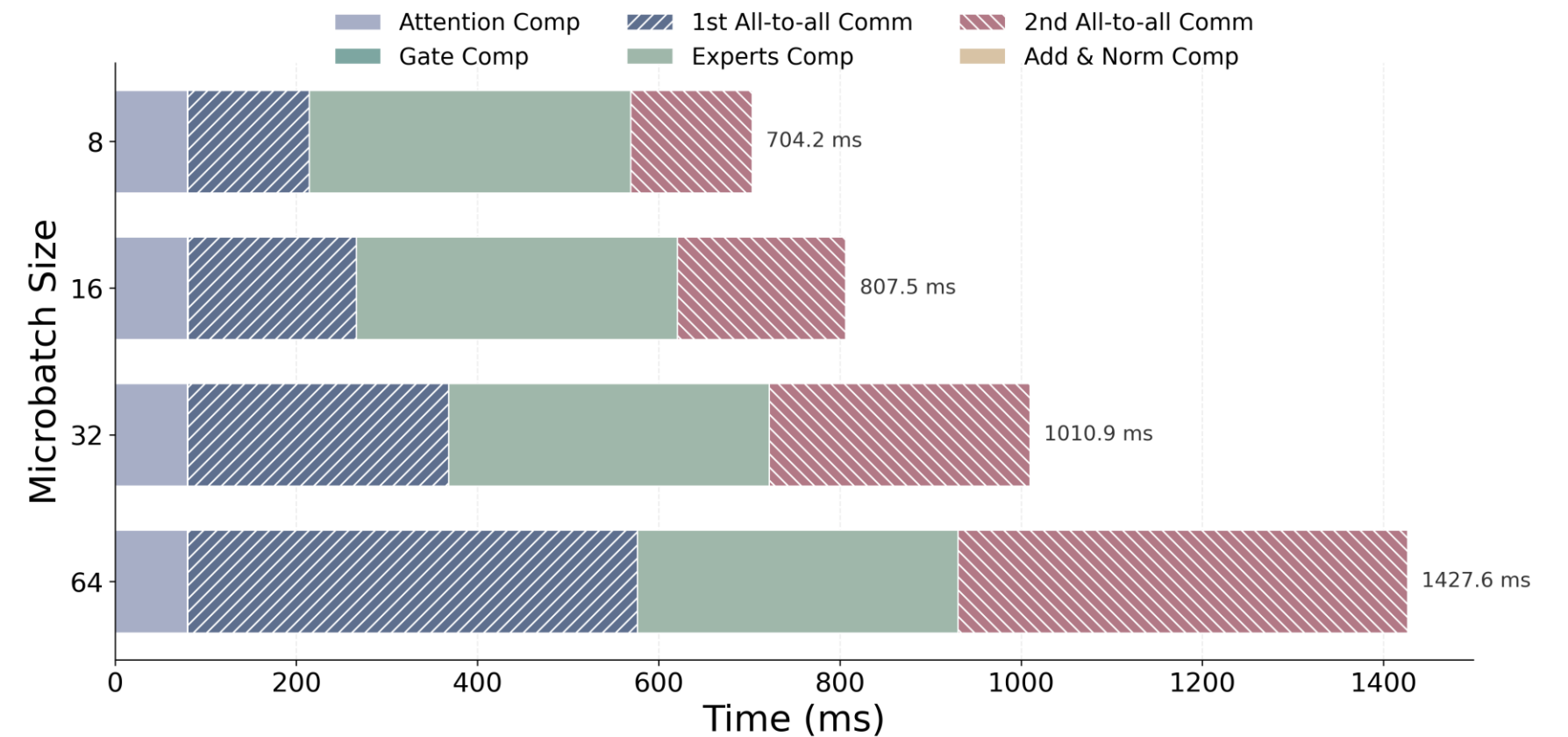
How long does it take for each kernels?

How does all-to-all communication occupy in the total iteration time?

Profile LLM MoE models



Mixtral 8x7B MoE Model single layer forward pass profiling
in 128 Nvidia A100 GPUs with 400Gbps network

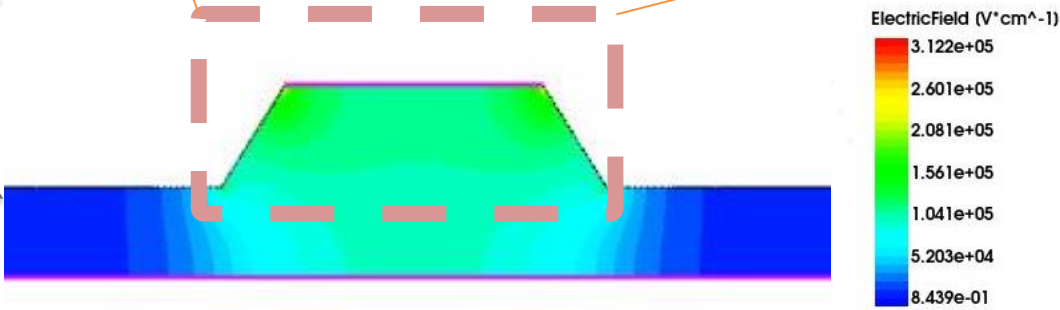
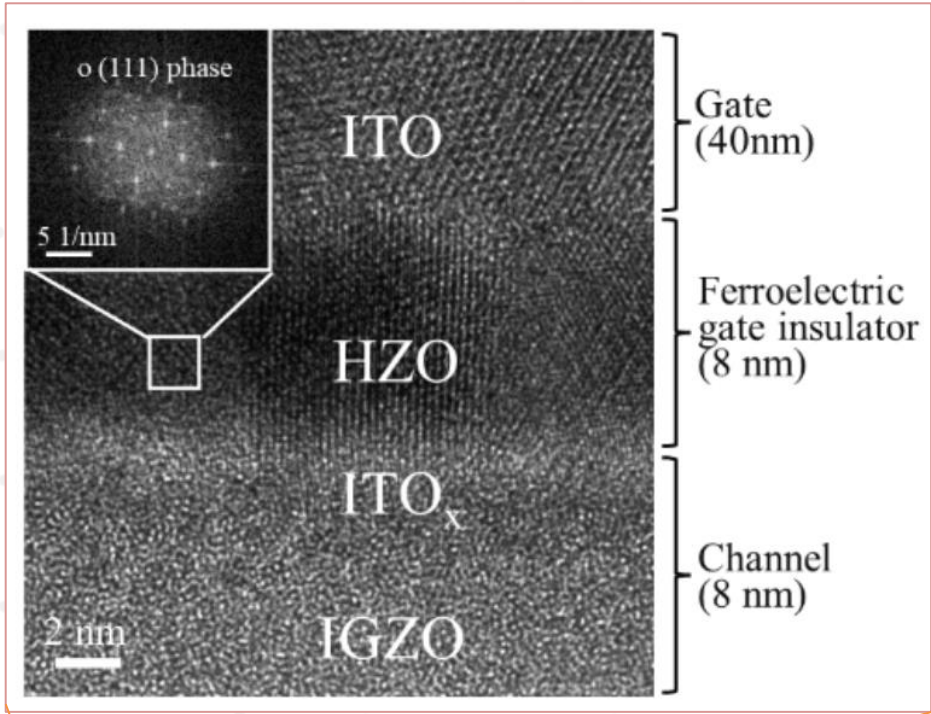


LLaMA-MoE 6.7B Model single layer forward pass profiling
in 64 Nvidia A100 GPUs with 400Gbps network

Expert Parallelism's All-to-All communications occupy 1/3 to 1/2 of the total training iteration time

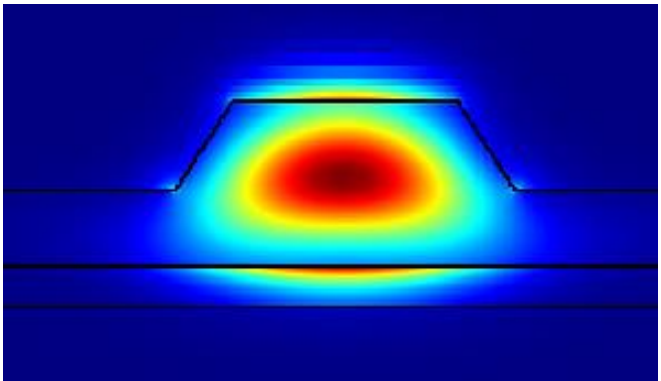
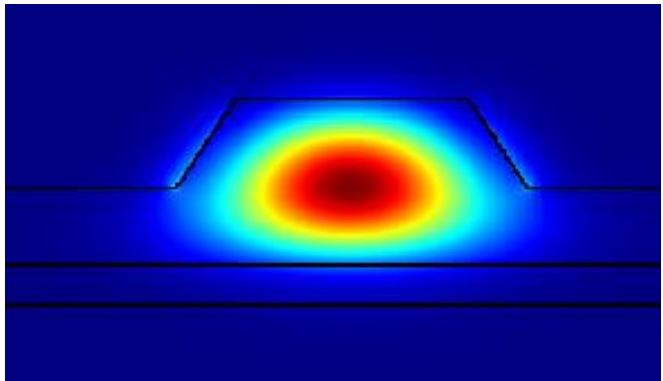
Ferroelectric MRR device simulation

Lithium Niobate
Waveguide



TCAD

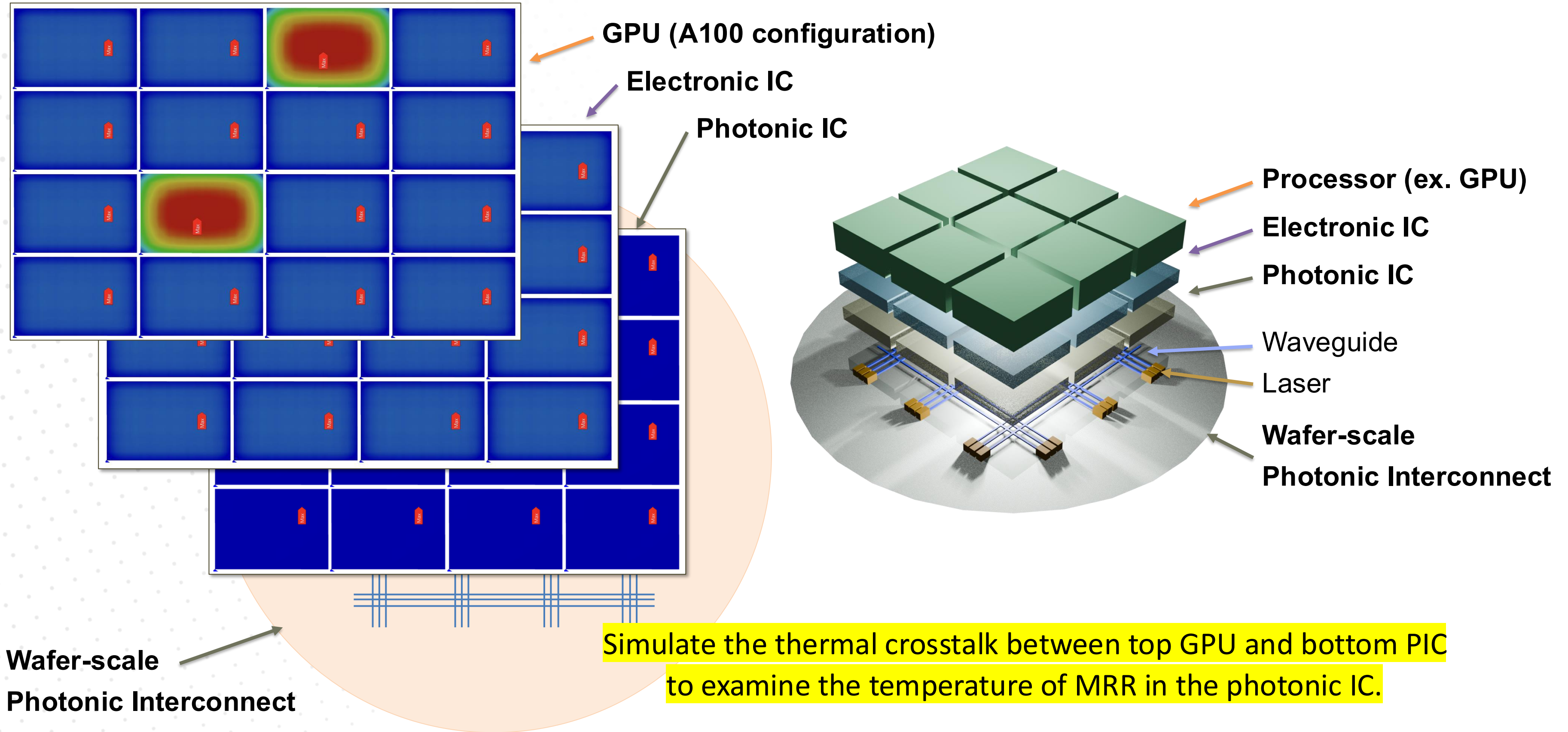
Electric field
Polarization
refractive index change



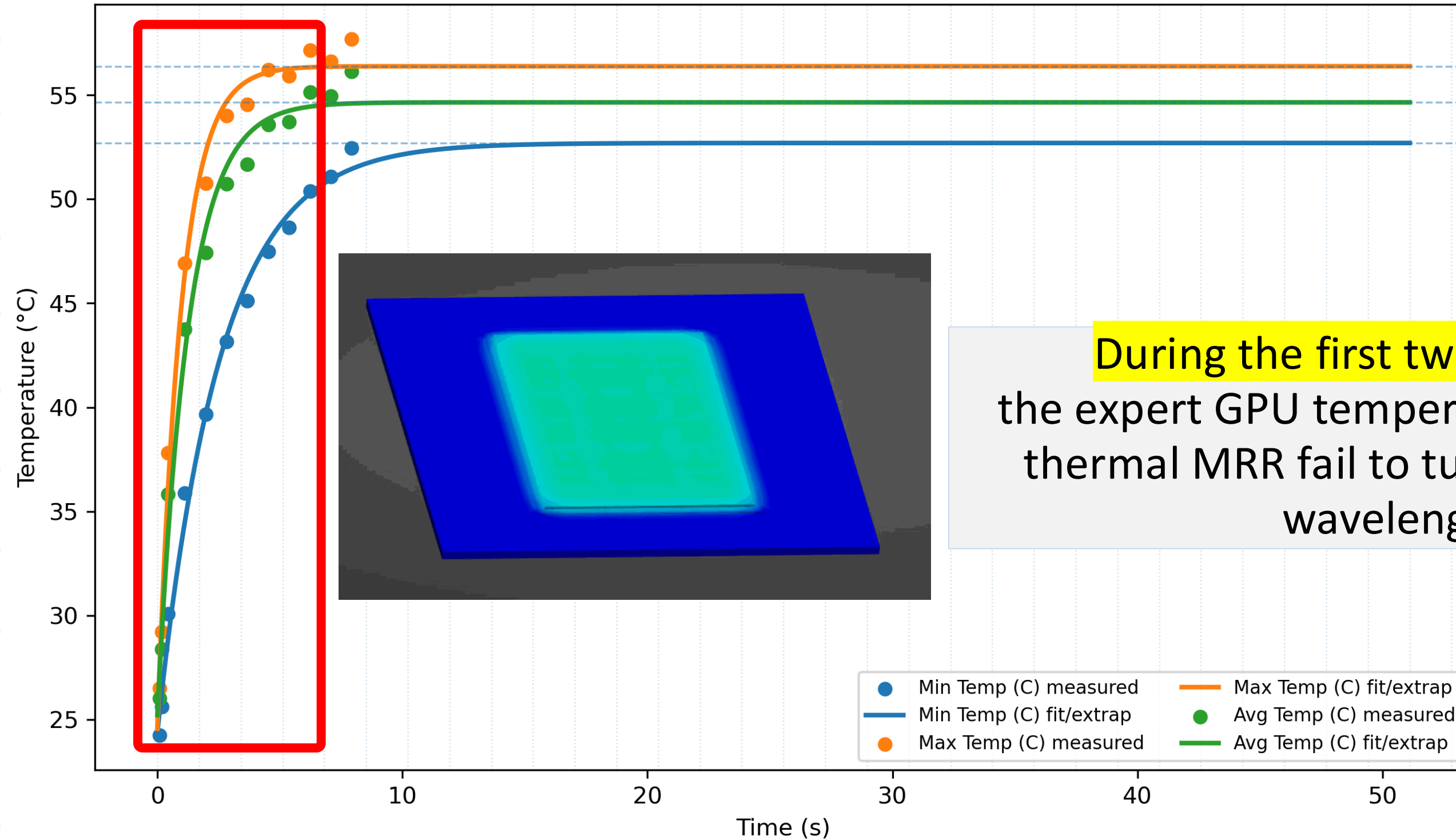
Ansys Lumerical

TE mode
TM mode

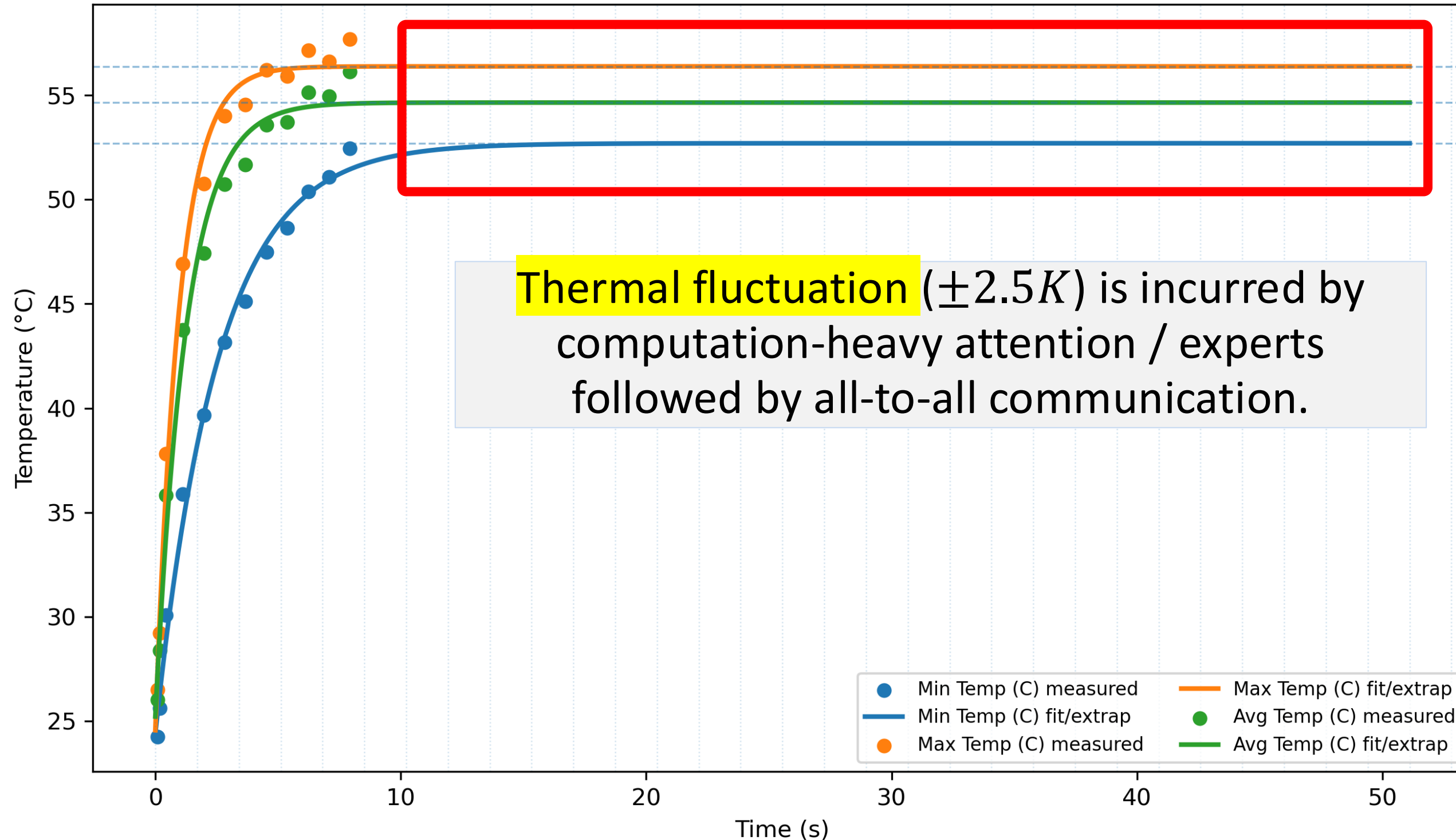
Thermal transient simulation with Ansys Thermal



Thermal transient response in expert GPU - MRR

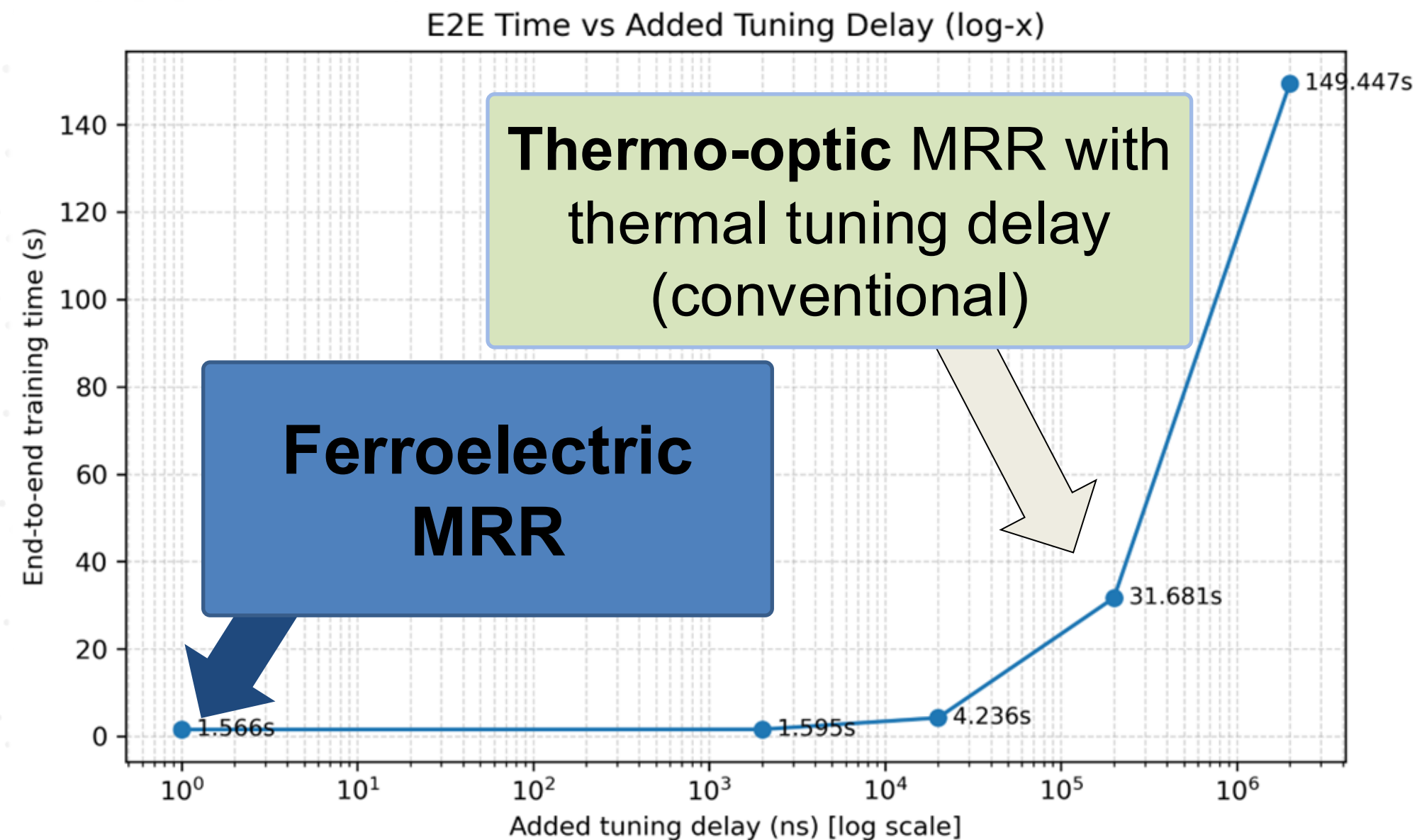


Thermal transient response in expert GPU - MRR



Therefore, thermal MRRs (require $\pm 0.1K$) takes longer all-to-all communication time as it require relaxation time for GPU to reach steady-state and tune operating wavelength.

Training speed-ups for LLM MoE model



Sharp decrease in end-to-end training iteration time in LLM MoE model with ferroelectric MRRs.

Recap

WE TALKED ABOUT ..

GenAI LLM Era

LLM / Mixture-of-Experts
Distributed Training
All-to-All traffic pattern

Distributed Infrastructure for ML

Datacenter Infrastructure
Communication Hierarchy

Optical Interconnect Solutions

Optical Device Module

1. Co-Packaged Optics (CPO)
2. Optical Interposer

**Ferroelectric MRR in
Wafer/Panel-Scale
Optical Interconnect Platform**

Reference

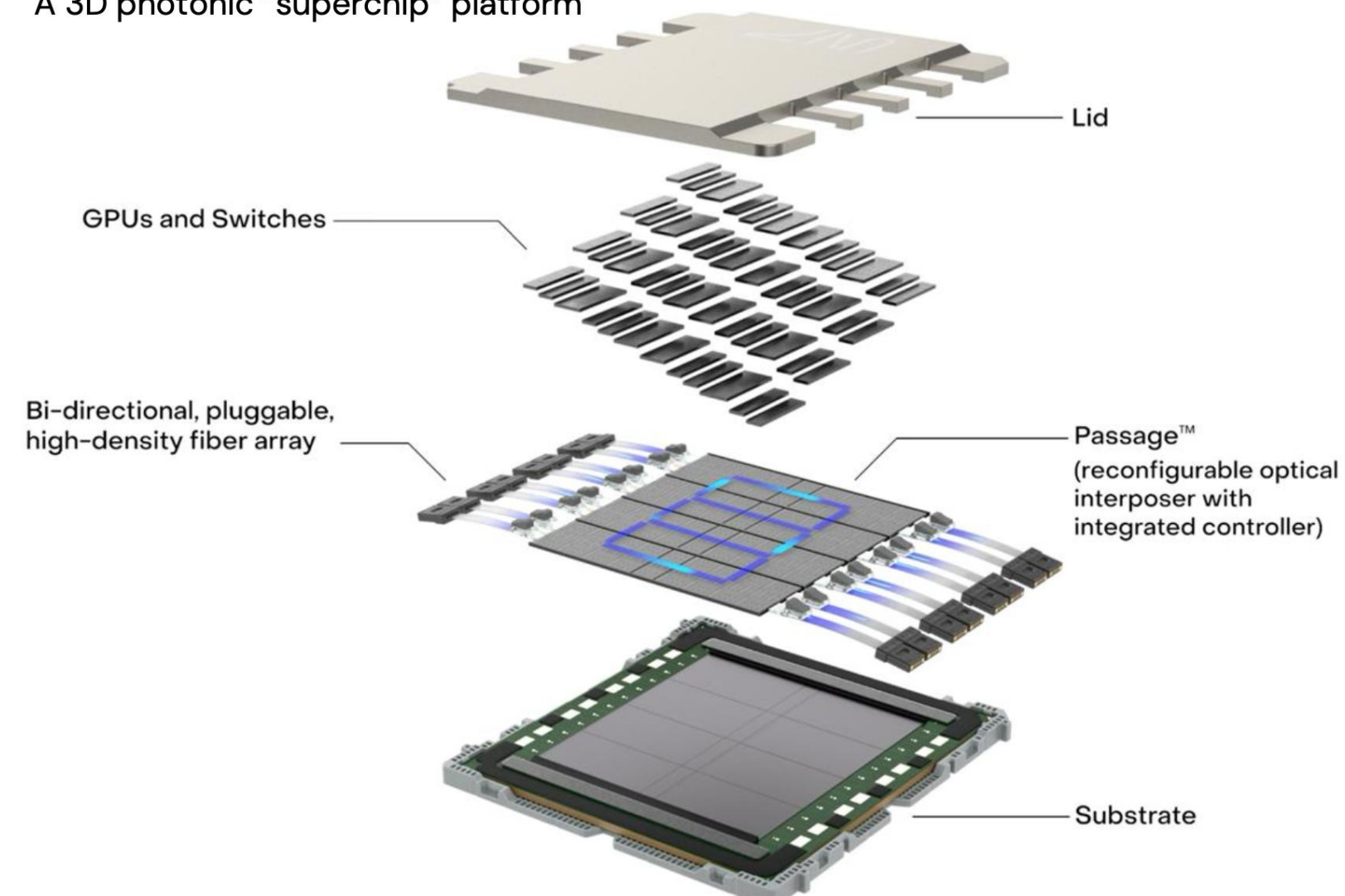
- ECE 8803 HML Lecture Note by Prof. Divya Mahajan in Georgia Institute of Technology
- HOTCHIPS 2025 AI Rack Tutorial, Frank Helms (AMD) “How AI workloads shape rack system architecture”
- HOTCHIPS 2025 AI Rack Tutorial, Darrin Valis (AMD) “Scaling Fabric Technologies”
- HOTCHIPS 2025 AI Rack Tutorial, John Norton (Nvidia) “Case study : Nvidia GB200NVL72”
- HOTCHIPS 2025 AI Rack Tutorial, Gilad Shainer (Nvidia) “Co-Packaged Silicon Photonics Switches for Gigawatt AI Factories”
- HOTCHIPS 2023, 2025 AI Rack Tutorial, Lightmatter
- HOTCHIPS 2025 AI Rack Tutorial, Celestial AI
- Zhongkai Yu et al., “Orders in Chaos: Enhancing Large-Scale MoE LLM Serving with Data Movement Forecasting”, *arXiv* (2025)
- Xudong Liao et al., “Mixnet: A Runtime Reconfigurable Optical-Electrical Fabric for Distributed Mixture-of-Experts Training”, ACM/IEEE SIGCOMM (2025)
- John H. Lau et al., “Co-Packaged Optics—Heterogeneous Integration of Photonic Integrated Circuits and Electronic Integrated Circuits” (2025)

Thank you

Additional Slides

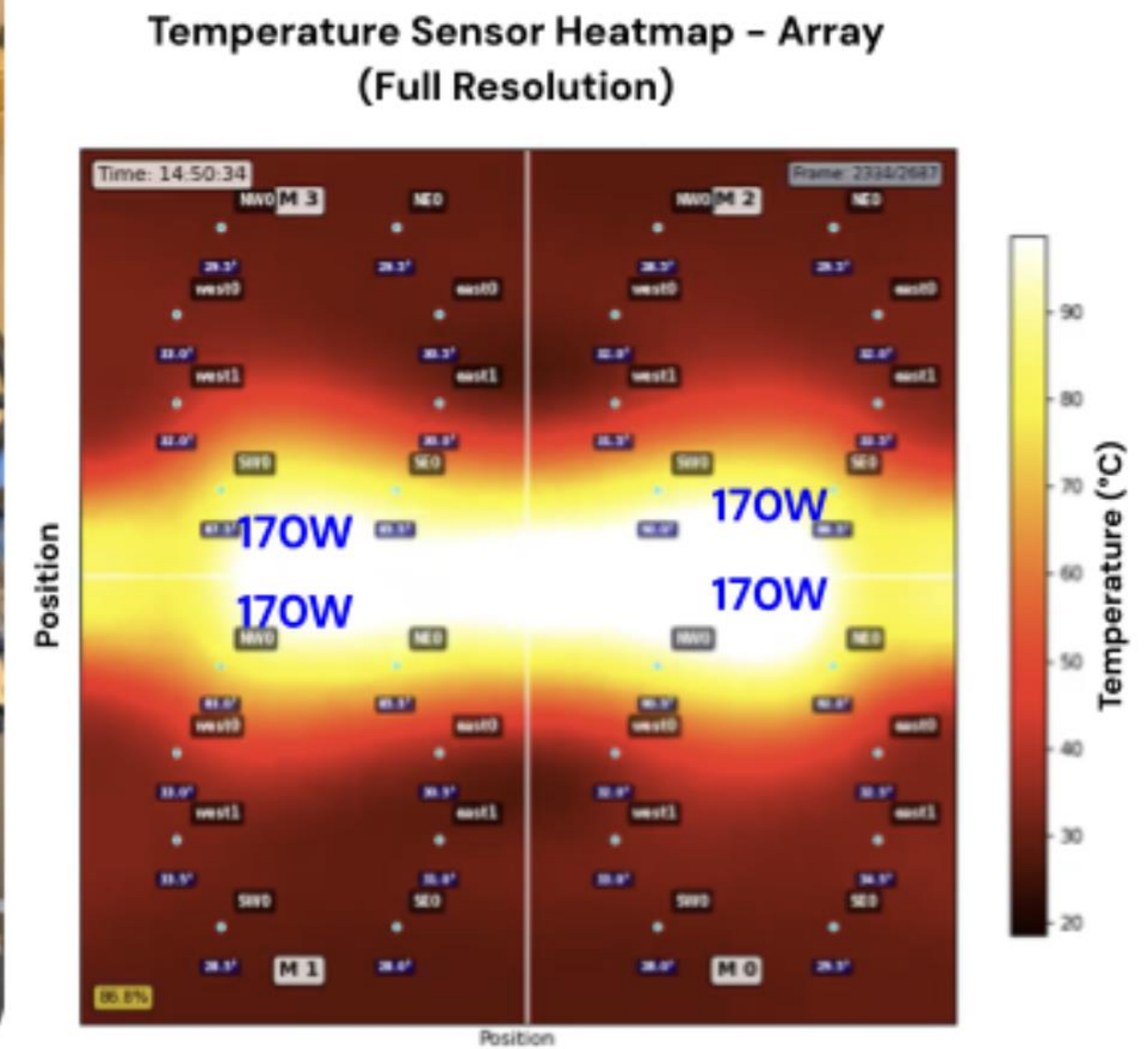
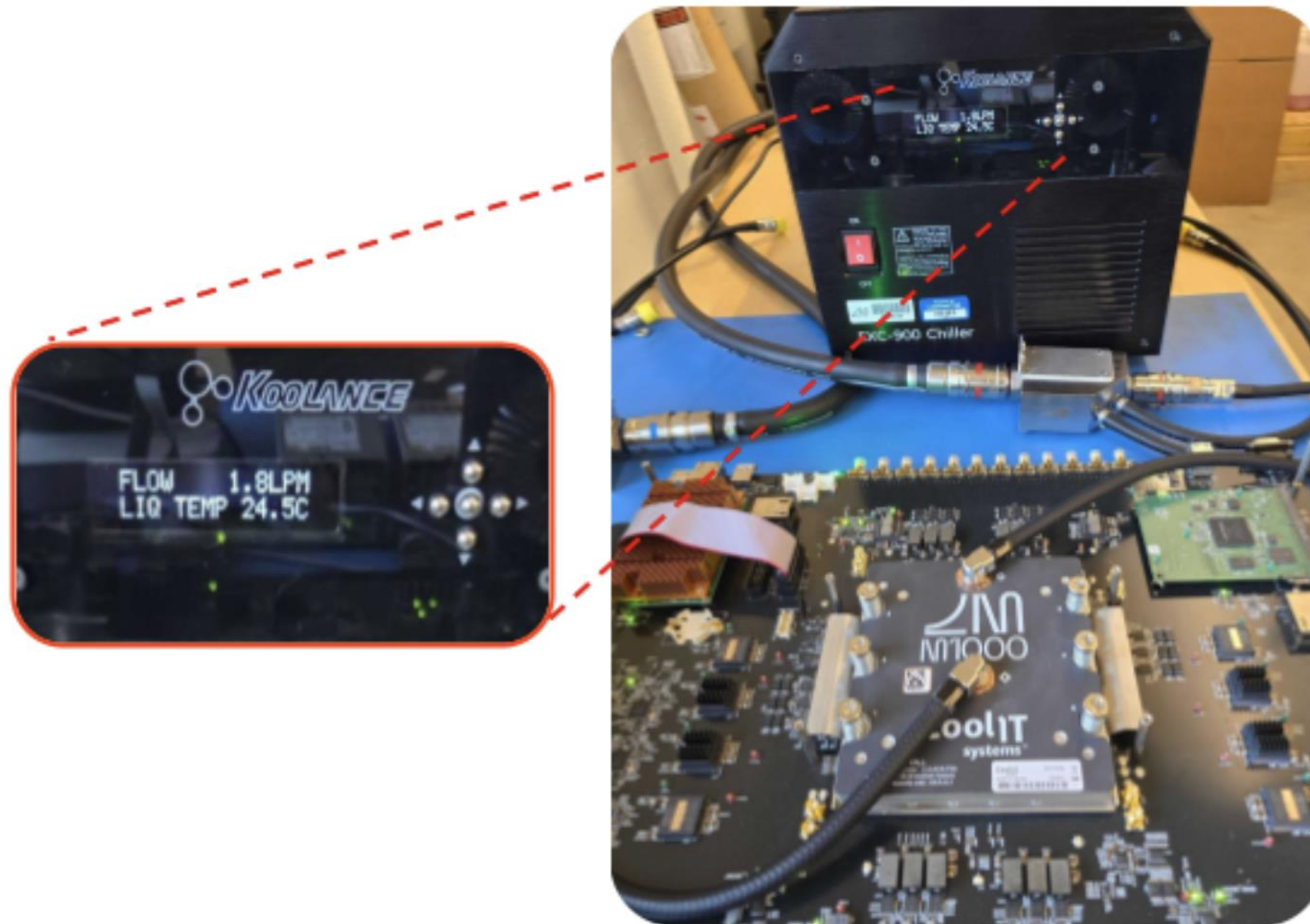
Passage M1000

A 3D photonic "superchip" platform



Lightmatter also manufacture wafer scale optical interconnect substrate

Additional Slides



However, Lightmatter only tested on steady-state thermal loads, not on transient thermal fluctuations with specific ML workloads.